3

Linear Dynamical Systems

This chapter reviews some basic system theoretical concepts. The notions of controllability, observability, stabilizability, and detectability are defined and various algebraic and geometric characterizations of these notions are summarized. Kalman canonical decomposition, pole placement, and observer theory are then introduced. The solutions of Lyapunov equations and their connections with system stability, controllability, and so on, are discussed. System interconnections and realizations, in particular the balanced realization, are studied in some detail. Finally, the concepts of system poles and zeros are introduced.

3.1 Descriptions of Linear Dynamical Systems

Let a finite dimensional linear time invariant (FDLTI) dynamical system be described by the following linear constant coefficient differential equations:

$$\dot{x} = Ax + Bu, \quad x(t_0) = x_0 \tag{3.1}$$

$$y = Cx + Du \tag{3.2}$$

where $x(t) \in \mathbb{R}^n$ is called the system *state*, $x(t_0)$ is called the *initial condition* of the system, $u(t) \in \mathbb{R}^m$ is called the system *input*, and $y(t) \in \mathbb{R}^p$ is the system *output*. The A, B, C, and D are appropriately dimensioned real constant matrices. A dynamical system with single input ($m = 1$) and single output ($p = 1$) is called a SISO (single input and single output) system, otherwise it is called MIMO (multiple input and multiple output) system. The corresponding transfer matrix from u to y is defined as

$$Y(s) = G(s)U(s)$$

where $U(s)$ and $Y(s)$ are the Laplace transform of $u(t)$ and $y(t)$ with zero initial condition ($x(0) = 0$). Hence, we have

$$G(s) = C(sI - A)^{-1}B + D.$$

Note that the system equations (3.1) and (3.2) can be written in a more compact matrix form:

$$\begin{bmatrix} \dot{x} \\ y \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix}.$$

To expedite calculations involving transfer matrices, the notation

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] := C(sI - A)^{-1}B + D$$

will be used. Other reasons for using this notation will be discussed in Chapter 10. Note that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

is a real block matrix, not a transfer function.

Now given the initial condition $x(t_0)$ and the input $u(t)$, the dynamical system response $x(t)$ and $y(t)$ for $t \geq t_0$ can be determined from the following formulas:

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)d\tau \tag{3.3}$$

$$y(t) = Cx(t) + Du(t). \tag{3.4}$$

In the case of $u(t) = 0$, $\forall t \geq t_0$, it is easy to see from the solution that for any $t_1 \geq t_0$ and $t \geq t_0$, we have

$$x(t) = e^{A(t-t_1)}x(t_1).$$

Therefore, the matrix function $\Phi(t, t_1) = e^{A(t-t_1)}$ acts as a transformation from one state to another, and thus $\Phi(t, t_1)$ is usually called the *state transition matrix*. Since the state of a linear system at one time can be obtained from the state at another through the transition matrix, we can assume without loss of generality that $t_0 = 0$. This will be assumed in the sequel.

The impulse matrix of the dynamical system is defined as

$$g(t) = \mathcal{L}^{-1}\{G(s)\} = Ce^{At}B1_+(t) + D\delta(t)$$

where $\delta(t)$ is the unit impulse and $1_+(t)$ is the unit step defined as

$$1_+(t) := \begin{cases} 1, & t \geq 0; \\ 0, & t < 0. \end{cases}$$

The input/output relationship (i.e., with zero initial state: $x_0 = 0$) can be described by the convolution equation

$$y(t) = (g * u)(t) := \int_{-\infty}^{\infty} g(t-\tau)u(\tau)d\tau = \int_{-\infty}^{t} g(t-\tau)u(\tau)d\tau.$$

3.2 Controllability and Observability

We now turn to some very important concepts in linear system theory.

Definition 3.1 The dynamical system described by the equation (3.1) or the pair (A, B) is said to be *controllable* if, for any initial state $x(0) = x_0$, $t_1 > 0$ and final state x_1, there exists a (piecewise continuous) input $u(\cdot)$ such that the solution of (3.1) satisfies $x(t_1) = x_1$. Otherwise, the system or the pair (A, B) is said to be *uncontrollable*.

The controllability (or observability introduced next) of a system can be verified through some algebraic or geometric criteria.

Theorem 3.1 *The following are equivalent:*

(i) (A, B) *is controllable.*

(ii) The matrix

$$W_c(t) := \int_0^t e^{A\tau}BB^*e^{A^*\tau}d\tau$$

is positive definite for any $t > 0$.

(iii) The controllability matrix

$$\mathcal{C} = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix}$$

has full row rank or, in other words, $\langle A \,| Im B\rangle := \sum_{i=1}^n Im(A^{i-1}B) = \mathbb{R}^n$.

(iv) The matrix $[A - \lambda I,\ B]$ *has full row rank for all* λ *in* $\mathbb{C}$.

(v) Let λ *and* x *be any eigenvalue and any corresponding left eigenvector of* A, *i.e.,* $x^*A = x^*\lambda$, *then* $x^*B \neq 0$.

(vi) The eigenvalues of $A+BF$ *can be freely assigned (with the restriction that complex eigenvalues are in conjugate pairs) by a suitable choice of* F.

Proof.

$(i) \Leftrightarrow (ii)$: Suppose $W_c(t_1) > 0$ for some $t_1 > 0$, and let the input be defined as

$$u(\tau) = -B^* e^{A^*(t_1-\tau)} W_c(t_1)^{-1}(e^{At_1}x_0 - x_1).$$

Then it is easy to verify using the formula in (3.3) that $x(t_1) = x_1$. Since x_1 is arbitrary, the pair (A, B) is controllable.

To show that the controllability of (A, B) implies that $W_c(t) > 0$ for any $t > 0$, assume that (A, B) is controllable but $W_c(t_1)$ is singular for some $t_1 > 0$. Since $e^{At}BB^*e^{A^*t} \geq 0$ for all t, there exists a real vector $0 \neq v \in \mathbb{R}^n$ such that

$$v^* e^{At} B = 0, \quad t \in [0, t_1].$$

Now let $x(t_1) = x_1 = 0$, and then from the solution (3.3), we have

$$0 = e^{At_1}x(0) + \int_0^{t_1} e^{A(t_1-\tau)}Bu(\tau)d\tau.$$

Pre-multiply the above equation by v^* to get

$$0 = v^* e^{At_1} x(0).$$

If we chose the initial state $x(0) = e^{-At_1}v$, then $v = 0$, and this is a contradiction. Hence, $W_c(t)$ can not be singular for any $t > 0$.

$(ii) \Leftrightarrow (iii)$: Suppose $W_c(t) > 0$ for all $t > 0$ (in fact, it can be shown that $W_c(t) > 0$ for all $t > 0$ iff, for some t_1, $W_c(t_1) > 0$) but the controllability matrix $\mathcal{C}$ does not have full row rank. Then there exists a $v \in \mathbb{R}^n$ such that

$$v^* A^i B = 0$$

for all $0 \leq i \leq n-1$. In fact, this equality holds for all $i \geq 0$ by the Cayley-Hamilton Theorem. Hence,

$$v^* e^{At} B = 0$$

for all t or, equivalently, $v^* W_c(t) = 0$ for all t; this is a contradiction, and hence, the controllability matrix $\mathcal{C}$ must be full row rank. Conversely, suppose $\mathcal{C}$ has full row rank but $W_c(t)$ is singular for some t_1. Then there exists a $0 \neq v \in \mathbb{R}^n$ such that $v^* e^{At} B = 0$ for all $t \in [0, t_1]$. Therefore, set $t = 0$, and we have

$$v^* B = 0.$$

Next, evaluate the i-th derivative of $v^* e^{At} B = 0$ at $t = 0$ to get

$$v^* A^i B = 0, \quad i > 0.$$

Hence, we have

$$v^* \begin{bmatrix} B & AB & A^2B & \ldots & A^{n-1}B \end{bmatrix} = 0$$

or, in other words, the controllability matrix $\mathcal{C}$ does not have full row rank. This is again a contradiction.

$(iii) \Rightarrow (iv)$: Suppose, on the contrary, that the matrix

$$\begin{bmatrix} A - \lambda I & B \end{bmatrix}$$

does not have full row rank for some $\lambda \in \mathbb{C}$. Then there exists a vector $x \in \mathbb{C}^n$ such that

$$x^* \begin{bmatrix} A - \lambda I & B \end{bmatrix} = 0$$

i.e., $x^* A = \lambda x^*$ and $x^* B = 0$. However, this will result in

$$x^* \begin{bmatrix} B & AB & \dots & A^{n-1}B \end{bmatrix} = \begin{bmatrix} x^*B & \lambda x^*B & \dots & \lambda^{n-1}x^*B \end{bmatrix} = 0$$

i.e., the controllability matrix $\mathcal{C}$ does not have full row rank, and this is a contradiction.

$(iv) \Rightarrow (v)$: This is obvious from the proof of $(iii) \Rightarrow (iv)$.

$(v) \Rightarrow (iii)$: We will again prove this by contradiction. Assume that (v) holds but rank $\mathcal{C} = k < n$. Then in section 3.3, we will show that there is a transformation T such that

$$TAT^{-1} = \begin{bmatrix} \bar{A}_c & \bar{A}_{12} \\ 0 & \bar{A}_{\bar{c}} \end{bmatrix} \quad TB = \begin{bmatrix} \bar{B}_c \\ 0 \end{bmatrix}$$

with $\bar{A}_{\bar{c}} \in \mathbb{R}^{(n-k)\times(n-k)}$. Let λ_1 and $x_{\bar{c}}$ be any eigenvalue and any corresponding left eigenvector of $\bar{A}_{\bar{c}}$, i.e., $x_{\bar{c}}^* \bar{A}_{\bar{c}} = \lambda_1 x_{\bar{c}}^*$. Then $x^*(TB) = 0$ and

$$x = \begin{bmatrix} 0 \\ x_{\bar{c}} \end{bmatrix}$$

is an eigenvector of TAT^{-1} corresponding to the eigenvalue λ_1, which implies that (TAT^{-1}, TB) is not controllable. This is a contradiction since similarity transformation does not change controllability. Hence, the proof is completed.

$(vi) \Rightarrow (i)$: This follows the same arguments as in the proof of $(v) \Rightarrow (iii)$: assume that (vi) holds but (A, B) is uncontrollable. Then, there is a decomposition so that some subsystems are not affected by the control, but this contradicts the condition (vi).

$(i) \Rightarrow (vi)$: This will be clear in section 3.4. In that section, we will explicitly construct a matrix F so that the eigenvalues of $A + BF$ are in the desired locations.

□

Definition 3.2 An unforced dynamical system $\dot{x} = Ax$ is said to be *stable* if all the eigenvalues of A are in the open left half plane, i.e., $\text{Re}\lambda(A) < 0$. A matrix A with such a property is said to be stable or Hurwitz.

Definition 3.3 The dynamical system (3.1), or the pair (A, B), is said to be *stabilizable* if there exists a state feedback $u = Fx$ such that the system is stable, i.e., $A + BF$ is stable.

Therefore, it is more appropriate to call this stabilizability the *state feedback stabilizability* to differentiate it from the *output feedback stabilizability* defined later.

The following theorem is a consequence of Theorem 3.1.

Theorem 3.2 *The following are equivalent:*

(i) (A, B) *is stabilizable.*

(ii) The matrix $[A - \lambda I, B]$ *has full row rank for all* $Re\lambda \geq 0$.

(iii) For all λ *and* x *such that* $x^* A = x^* \lambda$ *and* $Re\lambda \geq 0$, $x^* B \neq 0$.

(iv) There exists a matrix F *such that* $A + BF$ *is Hurwitz.*

We now consider the dual notions of observability and detectability of the system described by equations (3.1) and (3.2).

Definition 3.4 The dynamical system described by the equations (3.1) and (3.2) or by the pair (C, A) is said to be *observable* if, for any $t_1 > 0$, the initial state $x(0) = x_0$ can be determined from the time history of the input $u(t)$ and the output $y(t)$ in the interval of $[0, t_1]$. Otherwise, the system, or (C, A), is said to be *unobservable*.

Theorem 3.3 *The following are equivalent:*

(i) (C, A) *is observable.*

(ii) The matrix

$$W_o(t) := \int_0^t e^{A^* \tau} C^* C e^{A\tau} d\tau$$

is positive definite for any $t > 0$.

(iii) The observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has full column rank or $\bigcap_{i=1}^{n} Ker(CA^{i-1}) = 0$.

(iv) The matrix $\begin{bmatrix} A - \lambda I \\ C \end{bmatrix}$ *has full column rank for all* λ *in* $\mathbb{C}$.

(v) *Let λ and y be any eigenvalue and any corresponding right eigenvector of A, i.e., $Ay = \lambda y$, then $Cy \neq 0$.*

(vi) *The eigenvalues of $A+LC$ can be freely assigned (with the restriction that complex eigenvalues are in conjugate pairs) by a suitable choice of L.*

(vii) *(A^*, C^*) is controllable.*

Proof. First, we will show the equivalence between conditions (i) and (iii). Once this is done, the rest will follow by the duality or condition (vii).

(i) $\Leftarrow$ *(iii)*: Note that given the input $u(t)$ and the initial condition x_0, the output in the time interval $[0, t_1]$ is given by

$$y(t) = Ce^{At}x(0) + \int_0^t Ce^{A(t-\tau)}Bu(\tau)d\tau + Du(t).$$

Since $y(t)$ and $u(t)$ are known, there is no loss of generality in assuming $u(t) = 0, \forall t$. Hence,

$$y(t) = Ce^{At}x(0), \quad t \in [0, t_1].$$

From this equation, we have

$$\begin{bmatrix} y(0) \\ \dot{y}(0) \\ \vdots \\ y^{(n-1)}(0) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} x(0)$$

where $y^{(i)}$ stands for the i-th derivative of y. Since the observability matrix $\mathcal{O}$ has full column rank, there is a unique solution $x(0)$ in the above equation. This completes the proof.

(i) $\Rightarrow$ *(iii)*: This will be proven by contradiction. Assume that (C, A) is observable but that the observability matrix does not have full column rank, i.e., there is a vector x_0 such that $\mathcal{O}x_0 = 0$ or equivalently $CA^i x_0 = 0$, $\forall i \geq 0$ by the Cayley-Hamilton Theorem. Now suppose the initial state $x(0) = x_0$, then $y(t) = Ce^{At}x(0) = 0$. This implies that the system is not observable since $x(0)$ cannot be determined from $y(t) \equiv 0$.

$\square$

Definition 3.5 The system, or the pair (C, A), is *detectable* if $A + LC$ is stable for some L.

Theorem 3.4 *The following are equivalent:*

(i) (C, A) *is detectable.*

(ii) The matrix $\begin{bmatrix} A - \lambda I \\ C \end{bmatrix}$ *has full column rank for all* $Re\lambda \geq 0$.

(iii) For all λ *and* x *such that* $Ax = \lambda x$ *and* $Re\lambda \geq 0$, $Cx \neq 0$.

(iv) There exists a matrix L *such that* $A + LC$ *is Hurwitz.*

(v) (A^*, C^*) *is stabilizable.*

The conditions (iv) and (v) of Theorem 3.1 and Theorem 3.3 and the conditions (ii) and (iii) of Theorem 3.2 and Theorem 3.4 are often called Popov-Belevitch-Hautus (PBH) tests. In particular, the following definitions of modal controllability and observability are often useful.

Definition 3.6 Let λ be an eigenvalue of A or, equivalently, a mode of the system. Then the mode λ is said to be controllable (observable) if $x^*B \neq 0$ ($Cx \neq 0$) for *all* left (right) eigenvectors of A associated with λ, i.e., $x^*A = \lambda x^*$ ($Ax = \lambda x$) and $0 \neq x \in \mathbb{C}^n$. Otherwise, the mode is said to be uncontrollable (unobservable).

It follows that a system is controllable (observable) if and only if every mode is controllable (observable). Similarly, a system is stabilizable (detectable) if and only if every unstable mode is controllable (observable).

For example, consider the following 4th order system:

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{cccc|c} \lambda_1 & 1 & 0 & 0 & 0 \\ 0 & \lambda_1 & 1 & 0 & 1 \\ 0 & 0 & \lambda_1 & 0 & \alpha \\ 0 & 0 & 0 & \lambda_2 & 1 \\ \hline 1 & 0 & 0 & \beta & 0 \end{array}\right]$$

with $\lambda_1 \neq \lambda_2$. Then, the mode λ_1 is not controllable if $\alpha = 0$, and λ_2 is not observable if $\beta = 0$. Note that if $\lambda_1 = \lambda_2$, the system is uncontrollable and unobservable for any α and β since in that case, both

$$x_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad x_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

are the left eigenvectors of A corresponding to λ_1. Hence any linear combination of x_1 and x_2 is still an eigenvector of A corresponding to λ_1. In particular, let $x = x_1 - \alpha x_2$, then $x^*B = 0$, and as a result, the system is not controllable. Similar arguments can be applied to check observability. However, if the B matrix is changed into a 4×2 matrix with the last two rows independent of each other, then the system is controllable even if $\lambda_1 = \lambda_2$. For example, the reader may easily verify that the system with

$$B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ \alpha & 1 \\ 1 & 0 \end{bmatrix}$$

is controllable for any α.

In general, for a system given in the Jordan canonical form, the controllability and observability can be concluded by inspection. The interested reader may easily derive some explicit conditions for the controllability and observability by using Jordan canonical form and the tests (iv) and (v) of Theorem 3.1 and Theorem 3.3.

3.3 Kalman Canonical Decomposition

There are usually many different coordinate systems to describe a dynamical system. For example, consider a simple pendulum, the motion of the pendulum can be uniquely determined either in terms of the angle of the string attached to the pendulum or in terms of the vertical displacement of the pendulum. However, in most cases, the angular displacement is a more natural description than the vertical displacement in spite of the fact that they both describe the same dynamical system. This is true for most physical dynamical systems. On the other hand, although some coordinates may not be natural descriptions of a physical dynamical system, they may make the system analysis and synthesis much easier.

In general, let $T \in \mathbb{R}^{n\times n}$ be a nonsingular matrix and define

$$\bar{x} = Tx.$$

Then the original dynamical system equations (3.1) and (3.2) become

$$\begin{aligned} \dot{\bar{x}} &= TAT^{-1}\bar{x} + TBu \\ y &= CT^{-1}\bar{x} + Du. \end{aligned}$$

These equations represent the same dynamical system for any nonsingular matrix T, and hence, we can regard these representations as equivalent. It is easy to see that the input/output transfer matrix is not changed under the coordinate transformation, i.e.,

$$G(s) = C(sI - A)^{-1}B + D = CT^{-1}(sI - TAT^{-1})^{-1}TB + D.$$

In this section, we will consider the system structure decomposition using coordinate transformation if the system is not completely controllable and/or is not completely observable. To begin with, let us consider further the dynamical systems related by a similarity transformation:

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \longmapsto \left[\begin{array}{c|c} \bar{A} & \bar{B} \\ \hline \bar{C} & \bar{D} \end{array}\right] = \left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right].$$

The controllability and observability matrices are related by

$$\bar{\mathcal{C}} = T\mathcal{C} \quad \bar{\mathcal{O}} = \mathcal{O}T^{-1}.$$

Moreover, the following theorem follows easily from the PBH tests or from the above relations.

Theorem 3.5 *The controllability (or stabilizability) and observability (or detectability) are invariant under similarity transformations.*

Using this fact, we can now show the following theorem.

Theorem 3.6 *If the controllability matrix $\mathcal{C}$ has rank $k_1 < n$, then there exists a similarity transformation*

$$\bar{x} = \begin{bmatrix} \bar{x}_c \\ \bar{x}_{\bar{c}} \end{bmatrix} = Tx$$

such that

$$\begin{aligned} \begin{bmatrix} \dot{\bar{x}}_c \\ \dot{\bar{x}}_{\bar{c}} \end{bmatrix} &= \begin{bmatrix} \bar{A}_c & \bar{A}_{12} \\ 0 & \bar{A}_{\bar{c}} \end{bmatrix} \begin{bmatrix} \bar{x}_c \\ \bar{x}_{\bar{c}} \end{bmatrix} + \begin{bmatrix} \bar{B}_c \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} \bar{C}_c & \bar{C}_{\bar{c}} \end{bmatrix} \begin{bmatrix} \bar{x}_c \\ \bar{x}_{\bar{c}} \end{bmatrix} + Du \end{aligned}$$

with $\bar{A}_c \in \mathbb{C}^{k_1 \times k_1}$ and $(\bar{A}_c, \bar{B}_c)$ controllable. Moreover,

$$G(s) = C(sI - A)^{-1}B + D = \bar{C}_c(sI - \bar{A}_c)^{-1}\bar{B}_c + D.$$

Proof. Since $\operatorname{rank} \mathcal{C} = k_1 < n$, the pair (A, B) is not controllable. Now let $q_1, q_2, \ldots, q_{k_1}$ be any linearly independent columns of $\mathcal{C}$. Let $q_i, i = k_1 + 1, \ldots, n$ be any $n - k_1$ linearly independent vectors such that the matrix

$$Q := \begin{bmatrix} q_1 & \cdots & q_{k_1} & q_{k_1+1} & \cdots & q_n \end{bmatrix}$$

is nonsingular. Define

$$T := Q^{-1}.$$

Then the transformation $\bar{x} = Tx$ will give the desired decomposition. To see that, note that for each $i = 1, 2, \ldots, k_1$, Aq_i can be written as a linear combination of $q_i, i = 1, 2, \ldots, k_1$ since Aq_i is a linear combination of the columns of $\mathcal{C}$ by the Cayley-Hamilton Theorem. Therefore, we have

$$\begin{aligned} AT^{-1} &= \begin{bmatrix} Aq_1 & \cdots & Aq_{k_1} & Aq_{k_1+1} & \cdots & Aq_n \end{bmatrix} \\ &= \begin{bmatrix} q_1 & \cdots & q_{k_1} & q_{k_1+1} & \cdots & q_n \end{bmatrix} \begin{bmatrix} \bar{A}_c & \bar{A}_{12} \\ 0 & \bar{A}_{\bar{c}} \end{bmatrix} \\ &= T^{-1} \begin{bmatrix} \bar{A}_c & \bar{A}_{12} \\ 0 & \bar{A}_{\bar{c}} \end{bmatrix} \end{aligned}$$

for some $k_1 \times k_1$ matrix $\bar{A}_c$. Similarly, each column of the matrix B is a linear combination of $q_i, i = 1, 2, \ldots, k_1$, hence

$$B = Q \begin{bmatrix} \bar{B}_c \\ 0 \end{bmatrix} = T^{-1} \begin{bmatrix} \bar{B}_c \\ 0 \end{bmatrix}$$

for some $\bar{B}_c \in \mathbb{C}^{k_1 \times m}$.

To show that $(\bar{A}_c, \bar{B}_c)$ is controllable, note that rank $\mathcal{C} = k_1$ and

$$\mathcal{C} = T^{-1} \begin{bmatrix} \bar{B}_c & \bar{A}_c \bar{B}_c & \cdots & \bar{A}_c^{k_1 - 1} \bar{B}_c & \cdots & \bar{A}_c^{n-1} \bar{B}_c \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}.$$

Since, for each $j \geq k_1$, $\bar{A}_c^j$ is a linear combination of $\bar{A}_c^i, i = 0, 1, \ldots, (k_1 - 1)$ by Cayley-Hamilton theorem, we have

$$\text{rank} \begin{bmatrix} \bar{B}_c & \bar{A}_c \bar{B}_c & \cdots & \bar{A}_c^{k_1 - 1} \bar{B}_c \end{bmatrix} = k_1,$$

i.e., $(\bar{A}_c, \bar{B}_c)$ is controllable. $\square$

A numerically reliable way to find a such transformation T is to use QR factorization. For example, if the controllability matrix $\mathcal{C}$ has the QR factorization $QR = \mathcal{C}$, then $T = Q^{-1}$.

Corollary 3.7 *If the system is stabilizable and the controllability matrix $\mathcal{C}$ has rank $k_1 < n$, then there exists a similarity transformation T such that*

$$\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right] = \left[\begin{array}{cc|c} \bar{A}_c & \bar{A}_{12} & \bar{B}_c \\ 0 & \bar{A}_{\bar{c}} & 0 \\ \hline \bar{C}_c & \bar{C}_{\bar{c}} & D \end{array}\right]$$

with $\bar{A}_c \in \mathbb{C}^{k_1 \times k_1}$, $(\bar{A}_c, \bar{B}_c)$ controllable and with $\bar{A}_{\bar{c}}$ stable.

Hence, the state space $\bar{x}$ is partitioned into two orthogonal subspaces

$$\left\{ \begin{bmatrix} \bar{x}_c \\ 0 \end{bmatrix} \right\} \quad \text{and} \quad \left\{ \begin{bmatrix} 0 \\ \bar{x}_{\bar{c}} \end{bmatrix} \right\}$$

with the first subspace controllable from the input and second completely uncontrollable from the input (i.e., the state $\bar{x}_{\bar{c}}$ are not affected by the control u). Write these subspaces in terms of the original coordinates x, we have

$$x = T^{-1} \bar{x} = \begin{bmatrix} q_1 & \cdots & q_{k_1} & q_{k_1+1} & \cdots & q_n \end{bmatrix} \begin{bmatrix} \bar{x}_c \\ \bar{x}_{\bar{c}} \end{bmatrix}.$$

So the controllable subspace is the span of $q_i, i = 1, \ldots, k_1$ or, equivalently, Im $\mathcal{C}$. On the other hand, the uncontrollable subspace is given by the complement of the controllable subspace.

By duality, we have the following decomposition if the system is not completely observable.

Theorem 3.8 *If the observability matrix $\mathcal{O}$ has rank $k_2 < n$, then there exists a similarity transformation T such that*

$$\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right] = \left[\begin{array}{cc|c} \bar{A}_o & 0 & \bar{B}_o \\ \bar{A}_{21} & \bar{A}_{\bar{o}} & \bar{B}_{\bar{o}} \\ \hline \bar{C}_o & 0 & D \end{array}\right]$$

with $\bar{A}_o \in \mathbb{C}^{k_2 \times k_2}$ and $(\bar{C}_o, \bar{A}_o)$ observable.

Corollary 3.9 *If the system is detectable and the observability matrix $\mathcal{C}$ has rank $k_2 < n$, then there exists a similarity transformation T such that*

$$\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right] = \left[\begin{array}{cc|c} \bar{A}_o & 0 & \bar{B}_o \\ \bar{A}_{21} & \bar{A}_{\bar{o}} & \bar{B}_{\bar{o}} \\ \hline \bar{C}_o & 0 & D \end{array}\right]$$

with $\bar{A}_o \in \mathbb{C}^{k_2 \times k_2}$, $(\bar{C}_o, \bar{A}_o)$ observable and with $\bar{A}_{\bar{o}}$ stable.

Similarly, we have

$$G(s) = C(sI - A)^{-1}B + D = \bar{C}_o(sI - \bar{A}_o)^{-1}\bar{B}_o + D.$$

Carefully combining the above two theorems, we get the following *Kalman Canonical Decomposition*. The proof is left to the reader as an exercise.

Theorem 3.10 *Let an LTI dynamical system be described by the equations (3.1) and (3.2). Then there exists a nonsingular coordinate transformation $\bar{x} = Tx$ such that*

$$\begin{aligned} \begin{bmatrix} \dot{\bar{x}}_{co} \\ \dot{\bar{x}}_{c\bar{o}} \\ \dot{\bar{x}}_{\bar{c}o} \\ \dot{\bar{x}}_{\bar{c}\bar{o}} \end{bmatrix} &= \begin{bmatrix} \bar{A}_{co} & 0 & \bar{A}_{13} & 0 \\ \bar{A}_{21} & \bar{A}_{c\bar{o}} & \bar{A}_{23} & \bar{A}_{24} \\ 0 & 0 & \bar{A}_{\bar{c}o} & 0 \\ 0 & 0 & \bar{A}_{43} & \bar{A}_{\bar{c}\bar{o}} \end{bmatrix} \begin{bmatrix} \bar{x}_{co} \\ \bar{x}_{c\bar{o}} \\ \bar{x}_{\bar{c}o} \\ \bar{x}_{\bar{c}\bar{o}} \end{bmatrix} + \begin{bmatrix} \bar{B}_{co} \\ \bar{B}_{c\bar{o}} \\ 0 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} \bar{C}_{co} & 0 & \bar{C}_{\bar{c}o} & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_{co} \\ \bar{x}_{c\bar{o}} \\ \bar{x}_{\bar{c}o} \\ \bar{x}_{\bar{c}\bar{o}} \end{bmatrix} + Du \end{aligned}$$

or equivalently

$$\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right] = \left[\begin{array}{cccc|c} \bar{A}_{co} & 0 & \bar{A}_{13} & 0 & \bar{B}_{co} \\ \bar{A}_{21} & \bar{A}_{c\bar{o}} & \bar{A}_{23} & \bar{A}_{24} & \bar{B}_{c\bar{o}} \\ 0 & 0 & \bar{A}_{\bar{c}o} & 0 & 0 \\ 0 & 0 & \bar{A}_{43} & \bar{A}_{\bar{c}\bar{o}} & 0 \\ \hline \bar{C}_{co} & 0 & \bar{C}_{\bar{c}o} & 0 & D \end{array}\right]$$

where the vector $\bar{x}_{co}$ is controllable and observable, $\bar{x}_{c\bar{o}}$ is controllable but unobservable, $\bar{x}_{\bar{c}o}$ is observable but uncontrollable, and $\bar{x}_{\bar{c}\bar{o}}$ is uncontrollable and unobservable. Moreover, the transfer matrix from u to y is given by

$$G(s) = \bar{C}_{co}(sI - \bar{A}_{co})^{-1}\bar{B}_{co} + D.$$

One important issue is that although the transfer matrix of a dynamical system

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

is equal to its controllable and observable part

$$\left[\begin{array}{c|c} \bar{A}_{co} & \bar{B}_{co} \\ \hline \bar{C}_{co} & D \end{array}\right]$$

their internal behaviors are very different. In other words, while their input/output behaviors are the same, their state space response with nonzero initial conditions are very different. This can be illustrated by the state space response for the simple system

$$\begin{aligned}
\begin{bmatrix} \dot{\bar{x}}_{co} \\ \dot{\bar{x}}_{c\bar{o}} \\ \dot{\bar{x}}_{\bar{c}o} \\ \dot{\bar{x}}_{\bar{c}\bar{o}} \end{bmatrix} &= \begin{bmatrix} \bar{A}_{co} & 0 & 0 & 0 \\ 0 & \bar{A}_{c\bar{o}} & 0 & 0 \\ 0 & 0 & \bar{A}_{\bar{c}o} & 0 \\ 0 & 0 & 0 & \bar{A}_{\bar{c}\bar{o}} \end{bmatrix} \begin{bmatrix} \bar{x}_{co} \\ \bar{x}_{c\bar{o}} \\ \bar{x}_{\bar{c}o} \\ \bar{x}_{\bar{c}\bar{o}} \end{bmatrix} + \begin{bmatrix} \bar{B}_{co} \\ \bar{B}_{c\bar{o}} \\ 0 \\ 0 \end{bmatrix} u \\
y &= \begin{bmatrix} \bar{C}_{co} & 0 & \bar{C}_{\bar{c}o} & 0 \end{bmatrix} \begin{bmatrix} \bar{x}_{co} \\ \bar{x}_{c\bar{o}} \\ \bar{x}_{\bar{c}o} \\ \bar{x}_{\bar{c}\bar{o}} \end{bmatrix}
\end{aligned}$$

with $\bar{x}_{co}$ controllable and observable, $\bar{x}_{c\bar{o}}$ controllable but unobservable, $\bar{x}_{\bar{c}o}$ observable but uncontrollable, and $\bar{x}_{\bar{c}\bar{o}}$ uncontrollable and unobservable.

The solution to this system is given by

$$\begin{aligned}
\begin{bmatrix} \bar{x}_{co}(t) \\ \bar{x}_{c\bar{o}}(t) \\ \bar{x}_{\bar{c}o}(t) \\ \bar{x}_{\bar{c}\bar{o}}(t) \end{bmatrix} &= \begin{bmatrix} e^{\bar{A}_{co}t}\bar{x}_{co}(0) + \int_0^t e^{\bar{A}_{co}(t-\tau)}\bar{B}_{co}u(\tau)d\tau \\ e^{\bar{A}_{c\bar{o}}t}\bar{x}_{c\bar{o}}(0) + \int_0^t e^{\bar{A}_{c\bar{o}}(t-\tau)}\bar{B}_{c\bar{o}}u(\tau)d\tau \\ e^{\bar{A}_{\bar{c}o}t}\bar{x}_{\bar{c}o}(0) \\ e^{\bar{A}_{\bar{c}\bar{o}}t}\bar{x}_{\bar{c}\bar{o}}(0) \end{bmatrix} \\
y(t) &= \bar{C}_{co}\bar{x}_{co}(t) + \bar{C}_{\bar{c}o}\bar{x}_{\bar{c}o};
\end{aligned}$$

note that $\bar{x}_{\bar{c}o}(t)$ and $\bar{x}_{\bar{c}\bar{o}}(t)$ are not affected by the input u, while $\bar{x}_{c\bar{o}}(t)$ and $\bar{x}_{\bar{c}\bar{o}}(t)$ do not show up in the output y. Moreover, if the initial condition is zero, i.e., $\bar{x}(0) = 0$, then the output

$$y(t) = \int_0^t \bar{C}_{co}e^{\bar{A}_{co}(t-\tau)}\bar{B}_{co}u(\tau)d\tau.$$

However, if the initial state is not zero, then the response $\bar{x}_{\bar{c}o}(t)$ will show up in the output. In particular, if $\bar{A}_{\bar{c}o}$ is not stable, then the output $y(t)$ will grow without bound. The problems with uncontrollable and/or unobservable unstable modes are even more profound than what we have mentioned. Since the states are the internal signals of the dynamical system, any unbounded internal signal will eventually destroy the system. On the other hand, since it is impossible to make the initial states *exactly* zero, any uncontrollable and/or unobservable unstable mode will result in unacceptable system behavior. This issue will be exploited further in section 3.7. In the next section, we will consider how to place the system poles to achieve desired closed-loop behavior if the system is controllable.

3.4 Pole Placement and Canonical Forms

Consider a MIMO dynamical system described by

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du, \end{aligned}$$

and let u be a state feedback control law given by

$$u = Fx + v.$$

This closed-loop system is as shown in Figure 3.1, and the closed-loop system equations are given by

$$\begin{aligned} \dot{x} &= (A + BF)x + Bv \\ y &= (C + DF)x + Dv. \end{aligned}$$

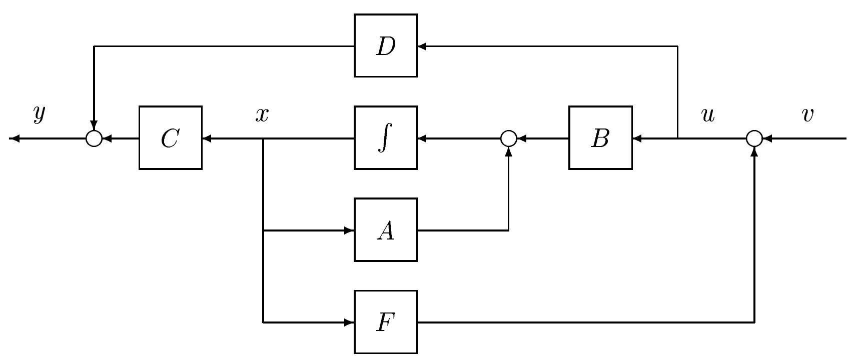


Figure 3.1: State Feedback

Then we have the following lemma in which the proof is simple and is left as an exercise to the reader.

Lemma 3.11 *Let F be a constant matrix with appropriate dimension; then (A, B) is controllable (stabilizable) if and only if $(A + BF, B)$ is controllable (stabilizable).*

However, the observability of the system may change under state feedback. For example, the following system

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 0 & 0 \\ \hline 1 & 0 & 0 \end{array}\right]$$

is controllable and observable. But with the state feedback

$$u = Fx = \begin{bmatrix} -1 & -1 \end{bmatrix} x,$$

the system becomes

$$\left[\begin{array}{c|c} A + BF & B \\ \hline C + DF & D \end{array}\right] = \left[\begin{array}{cc|c} 0 & 0 & 1 \\ 1 & 0 & 0 \\ \hline 1 & 0 & 0 \end{array}\right]$$

and is not completely observable.

The dual operation of the dynamical system by

$$\dot{x} = Ax + Bu \longmapsto \dot{x} = Ax + Bu + Ly$$

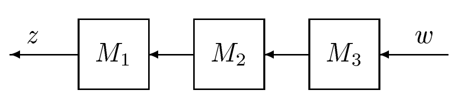
is called *output injection* which can be written as

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \longmapsto \left[\begin{array}{c|c} A + LC & B + LD \\ \hline C & D \end{array}\right].$$

By duality, the output injection does not change the system observability (detectability) but may change the system controllability (stabilizability).

Remark 3.1 We would like to call attention to the diagram and the signals flow convention used in this book. It may not be conventional to let signals flow from the right to the left, however, the reader will find that this representation is much more visually appealing than the traditional representation, and is consistent with the matrix manipulations. For example, the following diagram represents the multiplication of three matrices and will be very helpful in dealing with complicated systems:

$$z = M_1 M_2 M_3 w.$$



The conventional signal block diagrams, i.e., signals flowing from left to right, will also be used in this book. ♡

We now consider some special state space representations of the dynamical system described by equations (3.1) and (3.2). First, we will consider the systems with single inputs.

Assume that a single input and multiple output dynamical system is given by

$$G(s) = \left[\begin{array}{c|c} A & b \\ \hline C & d \end{array}\right], \quad b \in \mathbb{R}^n, \; C \in \mathbb{R}^{p\times n}, \; d \in \mathbb{R}^p$$

and assume that (A, b) is controllable. Let

$$\det(\lambda I - A) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_n,$$

and define

$$A_1 := \begin{bmatrix} -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \qquad b_1 := \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

and

$$\begin{aligned} \mathcal{C} &= \begin{bmatrix} b & Ab & \cdots & A^{n-1}b \end{bmatrix} \\ \mathcal{C}_1 &= \begin{bmatrix} b_1 & A_1b_1 & \cdots & A_1^{n-1}b_1 \end{bmatrix}. \end{aligned}$$

Then it is easy to verify that both $\mathcal{C}$ and $\mathcal{C}_1$ are nonsingular. Moreover, the transformation

$$T_c = \mathcal{C}_1\mathcal{C}^{-1}$$

will give the equivalent system representation

$$\left[\begin{array}{c|c} T_cAT_c^{-1} & T_cb \\ \hline CT_c^{-1} & d \end{array}\right] = \left[\begin{array}{c|c} A_1 & b_1 \\ \hline CT_c^{-1} & d \end{array}\right]$$

where

$$CT_c^{-1} = \begin{bmatrix} \beta_1 & \beta_2 & \cdots & \beta_{n-1} & \beta_n \end{bmatrix}$$

for some $\beta_i \in \mathbb{R}^p$. This state space representation is usually called *controllable canonical form* or *controller canonical form*. It is also easy to show that the transfer matrix is given by

$$G(s) = C(sI - A)^{-1}b + d = \frac{\beta_1 s^{n-1} + \beta_2 s^{n-2} + \cdots + \beta_{n-1}s + \beta_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n} + d,$$

which also shows that, given a column vector of transfer matrix $G(s)$, a state space representation of the transfer matrix can be obtained as above. The quadruple (A, b, C, d) is called a state space *realization* of the transfer matrix $G(s)$.

Now consider a dynamical system equation given by

$$\dot{x} = A_1 x + b_1 u$$

and a state feedback control law

$$u = Fx = \begin{bmatrix} f_1 & f_2 & \dots & f_{n-1} & f_n \end{bmatrix} x.$$

Then the closed-loop system is given by

$$\dot{x} = (A_1 + b_1 F)x$$

and $\det(\lambda I - (A_1 + b_1 F)) = \lambda^n + (a_1 - f_1)\lambda^{n-1} + \cdots + (a_n - f_n)$. It is clear that the zeros of $\det(\lambda I - (A_1 + b_1 F)$ can be made arbitrary for an appropriate choice of F provided that the complex zeros appear in conjugate pairs, thus showing that the eigenvalues of $A + bF$ can be freely assigned if (A, b) is controllable.

Dually, consider a multiple input and single output system

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline c & d \end{array}\right], \quad B \in \mathbb{R}^{n\times m},\ c^* \in \mathbb{R}^n,\ d^* \in \mathbb{R}^m,$$

and assume that (c, A) is observable; then there is a transformation T_o such that

$$\left[\begin{array}{c|c} T_o A T_o^{-1} & T_o B \\ \hline c T_o^{-1} & d \end{array}\right] = \left[\begin{array}{ccccc|c} -a_1 & 1 & 0 & \cdots & 0 & \eta_1 \\ -a_2 & 0 & 1 & \cdots & 0 & \eta_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 1 & \eta_{n-1} \\ -a_n & 0 & 0 & \cdots & 0 & \eta_n \\ \hline 1 & 0 & 0 & \cdots & 0 & d \end{array}\right], \quad \eta_i^* \in \mathbb{R}^m$$

and

$$G(s) = c(sI - A)^{-1}B + d = \frac{\eta_1 s^{n-1} + \eta_2 s^{n-2} + \cdots + \eta_{n-1}s + \eta_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n} + d.$$

This representation is called *observable canonical form* or *observer canonical form.*

Similarly, we can show that the eigenvalues of $A + Lc$ can be freely assigned if (c, A) is observable.

The pole placement problem for a multiple input system (A, B) can be converted into a simple single input pole placement problem. To describe the procedure, we need some preliminary results.

Lemma 3.12 *If an m input system pair (A, B) is controllable and if A is cyclic, then for almost all $v \in \mathbb{R}^m$, the single input pair (A, Bv) is controllable.*

Proof. Without loss of generality, assume that A is in the Jordan canonical form and that the matrix B is partitioned accordingly:

$$A = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{bmatrix} \quad B = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_k \end{bmatrix}$$

where J_i is in the form of

$$J_i = \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & \ddots & & \\ & & \ddots & \ddots & \\ & & & \lambda_i & 1 \\ & & & & \lambda_i \end{bmatrix}$$

and $\lambda_i \neq \lambda_j$ if $i \neq j$. By PBH test, the pair (A, B) is controllable if and only if, for each $i = 1, \ldots, k$, the last row of B_i is not zero. Let $b_i \in \mathbb{R}^m$ be the last row of B_i, and then we only need to show that, for almost all $v \in \mathbb{R}^m$, $b_i v \neq 0$ for each $i = 1, \ldots, k$ which is clear since for each i, the set $v \in \mathbb{R}^m$ such that $b_i v = 0$ has measure zero in $\mathbb{R}^m$ since $b_i \neq 0$. □

The cyclicity assumption in this theorem is essential. Without this assumption, the theorem does not hold. For example, the pair

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is controllable but there is no $v \in \mathbb{R}^2$ such that (A, Bv) is controllable since A is not cyclic.

Since a matrix A with distinct eigenvalues is cyclic, by definition we have the following lemma.

Lemma 3.13 *If (A, B) is controllable, then for almost any $K \in \mathbb{R}^{m \times n}$, all the eigenvalues of $A + BK$ are distinct and, consequently, $A + BK$ is cyclic.*

A proof can be found in Brasch and Pearson [1970], Davison [1968], and Heymann [1968].

Now it is evident that given a multiple input controllable pair (A, B), there is a matrix $K \in \mathbb{R}^{m \times n}$ and a vector $v \in \mathbb{R}^m$ such that $A + BK$ is cyclic and $(A + BK, Bv)$ is controllable. Moreover, from the pole placement results for the single input system, there is a matrix $f^* \in \mathbb{R}^n$ so that the eigenvalues of $(A + BK) + (Bv)f$ can be arbitrarily assigned. Hence, the eigenvalues of $A + BF$ can be arbitrarily assigned by choosing a state feedback in the form of

$$u = Fx = (K + vf)x.$$

A dual procedure can be applied for the output injection $A + LC$.

The canonical form for single input or single output system can also be generalized to multiple input and multiple output systems at the expense of notation. The interested reader may consult Kailath [1980] or Chen [1984].

If a system is not completely controllable, then the Kalman controllable decomposition can be applied first and the above procedure can be used to assign the system poles corresponding to the controllable subspace.

3.5 Observers and Observer-Based Controllers

We have shown in the last section that if a system is controllable and, furthermore, if the system states are available for feedback, then the system closed loop poles can be assigned arbitrarily through a constant feedback. However, in most practical applications, the system states are not completely accessible and all the designer knows are the output y and input u. Hence, the estimation of the system states from the given output information y and input u is often necessary to realize some specific design objectives. In this section, we consider such an estimation problem and the application of this state estimation in feedback control.

Consider a plant modeled by equations (3.1) and (3.2). An observer is a dynamical system with input of (u, y) and output of, say $\hat{x}$, which asymptotically estimates the state x. More precisely, a (linear) *observer* is a system such as

$$\begin{aligned} \dot{q} &= Mq + Nu + Hy \\ \hat{x} &= Qq + Ru + Sy \end{aligned}$$

so that $\hat{x}(t) - x(t) \to 0$ as $t \to \infty$ for all initial states $x(0)$, $q(0)$ and for every input $u(\cdot)$.

Theorem 3.14 *An observer exists iff (C, A) is detectable. Further, if (C, A) is detectable, then a full order Luenberger observer is given by*

$$\dot{q} = Aq + Bu + L(Cq + Du - y) \tag{3.5}$$

$$\hat{x} = q \tag{3.6}$$

where L is any matrix such that $A + LC$ is stable.

Proof. We first show that the detectability of (C, A) is sufficient for the existence of an observer. To that end, we only need to show that the so-called Luenberger observer defined in the theorem is indeed an observer. Note that equation (3.5) for q is a simulation of the equation for x, with an additional forcing term $L(Cq + Du - y)$, which is a gain times the output error. Equivalent equations are

$$\begin{aligned} \dot{q} &= (A + LC)q + Bu + LDu - Ly \\ \hat{x} &= q. \end{aligned}$$

These equations have the form allowed in the definition of an observer. Define the error, $e := \hat{x} - x$, and then simple algebra gives

$$\dot{e} = (A + LC)e;$$

therefore $e(t) \to 0$ as $t \to \infty$ for every $x(0)$, $q(0)$, and $u(.)$.

To show the converse, assume that (C, A) is not detectable. Take the initial state $x(0)$ in the undetectable subspace and consider a candidate observer:

$$\begin{aligned} \dot{q} &= Mq + Nu + Hy \\ \hat{x} &= Qq + Ru + Sy. \end{aligned}$$

Take $q(0) = 0$ and $u(t) \equiv 0$. Then the equations for x and the candidate observer are

$$\begin{aligned} \dot{x} &= Ax \\ \dot{q} &= Mq + HCx \\ \hat{x} &= Qq + SCx. \end{aligned}$$

Since an unobservable subspace is an A-invariant subspace containing $x(0)$, it follows that $x(t)$ is in the unobservable subspace for all $t \geq 0$. Hence, $Cx(t) = 0$ for all $t \geq 0$, and, consequently, $q(t) \equiv 0$ and $\hat{x}(t) \equiv 0$. However, for some $x(0)$ in the undetectable subspace, $x(t)$ does not converge to zero. Thus the candidate observer does not have the required property, and therefore, no observer exists. $\square$

The above Luenberger observer has dimension n, which is the dimension of the state x. It's possible to get an observer of lower dimension. The idea is this: since we can measure $y - Du = Cx$, we already know x modulo Ker C, so we only need to generate the part of x in Ker C. If C has full row rank and $p :=$ dim y, then the dimension of Ker C equals $n - p$, so we might suspect that we can get an observer of order $n - p$. This is true. Such an observer is called a "minimal order observer". We will not pursue this issue further here. The interested reader may consult Chen [1984].

Recall that, for a dynamical system described by the equations (3.1) and (3.2), if (A, B) is controllable and state x is available for feedback, then there is a state feedback $u = Fx$ such that the closed-loop poles of the system can be arbitrarily assigned. Similarly, if (C, A) is observable, then the system observer poles can be arbitrarily placed so that the state estimator $\hat{x}$ can be made to approach x arbitrarily fast. Now let us consider what will happen if the system states are not available for feedback so that the estimated state has to be used. Hence, the controller has the following dynamics:

$$\begin{aligned} \dot{\hat{x}} &= (A + LC)\hat{x} + Bu + LDu - Ly \\ u &= F\hat{x}. \end{aligned}$$

Then the total system state equations are given by

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A & BF \\ -LC & A+BF+LC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}.$$

Let $e := x - \hat{x}$, then the system equation becomes

$$\begin{bmatrix} \dot{e} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A+LC & 0 \\ -LC & A+BF \end{bmatrix} \begin{bmatrix} e \\ \hat{x} \end{bmatrix},$$

and the closed-loop poles consist of two parts: the poles resulting from state feedback $\sigma(A+BF)$ and the poles resulting from the state estimation $\sigma(A+LC)$. Now if (A,B) is controllable and (C,A) is observable, then there exist F and L such that the eigenvalues of $A+BF$ and $A+LC$ can be arbitrarily assigned. In particular, they can be made to be stable. Note that a slightly weaker result can also result even if (A,B) and (C,A) are only stabilizable and detectable.

The controller given above is called an observer-based controller and is denoted as

$$u = K(s)y$$

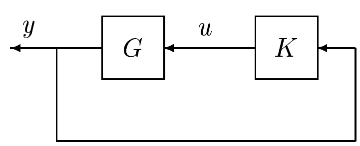
and

$$K(s) = \left[\begin{array}{c|c} A+BF+LC+LDF & -L \\ \hline F & 0 \end{array}\right].$$

Now denote the open loop plant by

$$G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right];$$

then the closed-loop feedback system is as shown below:



In general, if a system is stabilizable through feeding back the output y, then it is said to be *output feedback stabilizable.* It is clear from the above construction that a system is output feedback stabilizable if (A,B) is stabilizable and (C,A) is detectable. The converse is also true and will be shown in Chapter 12.

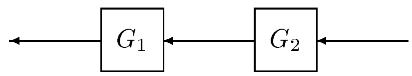
3.6 Operations on Systems

In this section, we present some facts about system interconnection. Since these proofs are straightforward, we will leave the details to the reader.

Suppose that G_1 and G_2 are two subsystems with state space representations:

$$G_1 = \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right] \qquad G_2 = \left[\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array}\right].$$

Then the series or cascade connection of these two subsystems is a system with the output of the second subsystem as the input of the first subsystem as shown in the following diagram:



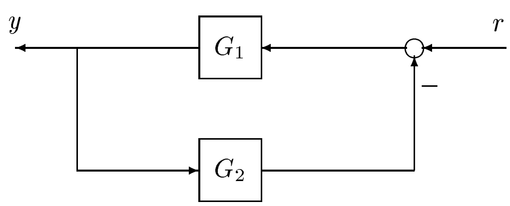
This operation in terms of the transfer matrices of the two subsystems is essentially the product of two transfer matrices. Hence, a representation for the cascaded system can be obtained as

$$\begin{aligned} G_1 G_2 &= \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right] \left[\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array}\right] \\ &= \left[\begin{array}{cc|c} A_1 & B_1 C_2 & B_1 D_2 \\ 0 & A_2 & B_2 \\ \hline C_1 & D_1 C_2 & D_1 D_2 \end{array}\right] = \left[\begin{array}{cc|c} A_2 & 0 & B_2 \\ B_1 C_2 & A_1 & B_1 D_2 \\ \hline D_1 C_2 & C_1 & D_1 D_2 \end{array}\right]. \end{aligned}$$

Similarly, the parallel connection or the addition of G_1 and G_2 can be obtained as

$$G_1 + G_2 = \left[\begin{array}{c|c} A_1 & B_1 \\ \hline C_1 & D_1 \end{array}\right] + \left[\begin{array}{c|c} A_2 & B_2 \\ \hline C_2 & D_2 \end{array}\right] = \left[\begin{array}{cc|c} A_1 & 0 & B_1 \\ 0 & A_2 & B_2 \\ \hline C_1 & C_2 & D_1 + D_2 \end{array}\right].$$

Next we consider a feedback connection of G_1 and G_2 as shown below:



Then the closed-loop transfer matrix from r to y is given by

$$T = \left[\begin{array}{cc|c} A_1 - B_1 D_2 R_{12}^{-1} C_1 & -B_1 R_{21}^{-1} C_2 & B_1 R_{21}^{-1} \\ B_2 R_{12}^{-1} C_1 & A_2 - B_2 D_1 R_{21}^{-1} C_2 & B_2 D_1 R_{21}^{-1} \\ \hline R_{12}^{-1} C_1 & -R_{12}^{-1} D_1 C_2 & D_1 R_{21}^{-1} \end{array}\right]$$

where $R_{12} = I + D_1 D_2$ and $R_{21} = I + D_2 D_1$. Note that these state space representations may not be necessarily controllable and/or observable even if the original subsystems G_1 and G_2 are.

For future reference, we shall also introduce the following definitions.

Definition 3.7 The *transpose* of a transfer matrix $G(s)$ or the *dual system* is defined as

$$G \longmapsto G^T(s) = B^*(sI - A^*)^{-1}C^* + D^*$$

or equivalently

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \longmapsto \left[\begin{array}{c|c} A^* & C^* \\ \hline B^* & D^* \end{array}\right].$$

Definition 3.8 The *conjugate* system of $G(s)$ is defined as

$$G \longmapsto G^{\sim}(s) := G^T(-s) = B^*(-sI - A^*)^{-1}C^* + D^*$$

or equivalently

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] \longmapsto \left[\begin{array}{c|c} -A^* & -C^* \\ \hline B^* & D^* \end{array}\right].$$

In particular, we have $G^*(j\omega) := [G(j\omega)]^* = G^{\sim}(j\omega)$.

Definition 3.9 A real rational matrix $\hat{G}(s)$ is called a *right (left) inverse* of a transfer matrix $G(s)$ if $G(s)\hat{G}(s) = I$ ($\hat{G}(s)G(s) = I$). Moreover, if $\hat{G}(s)$ is both a right inverse and a left inverse of $G(s)$, then it is simply called the inverse of $G(s)$.

Lemma 3.15 *Let $D^\dagger$ denote a right (left) inverse of D if D has full row (column) rank. Then*

$$G^\dagger = \left[\begin{array}{c|c} A - BD^\dagger C & -BD^\dagger \\ \hline D^\dagger C & D^\dagger \end{array}\right]$$

is a right (left) inverse of G.

Proof. The right inverse case will be proven and the left inverse case follows by duality. Suppose $DD^\dagger = I$. Then

$$\begin{aligned} GG^\dagger &= \left[\begin{array}{cc|c} A & BD^\dagger C & BD^\dagger \\ 0 & A - BD^\dagger C & -BD^\dagger \\ \hline C & DD^\dagger C & DD^\dagger \end{array}\right] \\ &= \left[\begin{array}{cc|c} A & BD^\dagger C & BD^\dagger \\ 0 & A - BD^\dagger C & -BD^\dagger \\ \hline C & C & I \end{array}\right]. \end{aligned}$$

Performing similarity transformation $T = \begin{bmatrix} I & I \\ 0 & I \end{bmatrix}$ on the above system yields

$$\begin{aligned} GG^\dagger &= \left[\begin{array}{cc|c} A & 0 & 0 \\ 0 & A - BD^\dagger C & -BD^\dagger \\ \hline C & 0 & I \end{array}\right] \\ &= I. \end{aligned}$$

□

3.7 State Space Realizations for Transfer Matrices

In some cases, the natural or convenient description for a dynamical system is in terms of transfer matrices. This occurs, for example, in some highly complex systems for which the analytic differential equations are too hard or too complex to write down. Hence, certain engineering approximation or identification has to be carried out; for example, input and output frequency responses are obtained from experiments so that some transfer matrix approximating the system dynamics is available. Since the state space computation is most convenient to implement on the computer, some appropriate state space representation for the resulting transfer matrix is necessary.

In general, assume that $G(s)$ is a real-rational transfer matrix which is *proper*. Then we call a state-space model (A, B, C, D) such that

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right],$$

a *realization* of $G(s)$.

We note that if the transfer matrix is either single input or single output, then the formulas in Section 3.4 can be used to obtain a controllable or observable realization. The realization for a general MIMO transfer matrix is more complicated and is the focus of this section.

Definition 3.10 A state space realization (A, B, C, D) of $G(s)$ is said to be a *minimal realization* of $G(s)$ if A has the smallest possible dimension.

Theorem 3.16 *A state space realization (A, B, C, D) of $G(s)$ is minimal if and only if (A, B) is controllable and (C, A) is observable.*

Proof. We shall first show that if (A, B, C, D) is a minimal realization of $G(s)$, then (A, B) must be controllable and (C, A) must be observable. Suppose, on the contrary, that (A, B) is not controllable and/or (C, A) is not observable. Then from Kalman decomposition, there is a smaller dimensioned controllable and observable state space realization that has the same transfer matrix; this contradicts the minimality assumption. Hence (A, B) must be controllable and (C, A) must be observable.

Next we show that if an n-th order realization (A, B, C, D) is controllable and observable, then it is minimal. But supposing it is not minimal, let (A_m, B_m, C_m, D) be a minimal realization of $G(s)$ with order $k < n$. Since

$$G(s) = C(sI - A)^{-1}B + D = C_m(sI - A_m)^{-1}B_m + D,$$

we have

$$CA^iB = C_mA_m^iB_m, \quad \forall i \geq 0.$$

This implies that

$$\mathcal{O}\mathcal{C} = \mathcal{O}_m\mathcal{C}_m \tag{3.7}$$

where $\mathcal{C}$ and $\mathcal{O}$ are the controllability and observability matrices of (A, B) and (C, A), respectively, and

$$\begin{aligned}
\mathcal{C}_m &:= \begin{bmatrix} B_m & A_m B_m & \cdots & A_m^{n-1} B_m \end{bmatrix} \\
\mathcal{O}_m &:= \begin{bmatrix} C_m \\ C_m A_m \\ \vdots \\ C_m A_m^{n-1} \end{bmatrix}.
\end{aligned}$$

By Sylvester's inequality,

$$\text{rank } \mathcal{C} + \text{rank } \mathcal{O} - n \leq \text{rank } (\mathcal{O}\mathcal{C}) \leq \min\{\text{rank } \mathcal{C}, \text{rank } \mathcal{O}\},$$

and, therefore, we have rank $(\mathcal{O}\mathcal{C}) = n$ since rank $\mathcal{C}$ = rank $\mathcal{O} = n$ by the controllability and observability assumptions. Similarly, since (A_m, B_m, C_m, D) is minimal, (A_m, B_m) is controllable and (C_m, A_m) is observable. Moreover,

$$\text{rank } \mathcal{O}_m \mathcal{C}_m = k < n,$$

which is a contradiction since rank $\mathcal{O}\mathcal{C}$ = rank $\mathcal{O}_m \mathcal{C}_m$ by equality (3.7). □

The following property of minimal realizations can also be verified, and this is left to the reader.

Theorem 3.17 *Let (A_1, B_1, C_1, D) and (A_2, B_2, C_2, D) be two minimal realizations of a real rational transfer matrix $G(s)$, and let $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{O}_1$, and $\mathcal{O}_2$ be the corresponding controllability and observability matrices, respectively. Then there exists a unique nonsingular T such that*

$$A_2 = T A_1 T^{-1}, \quad B_2 = T B_1, \quad C_2 = C_1 T^{-1}.$$

Furthermore, T can be specified as $T = (\mathcal{O}_2^* \mathcal{O}_2)^{-1} \mathcal{O}_2^* \mathcal{O}_1$ or $T^{-1} = \mathcal{C}_1 \mathcal{C}_2^* (\mathcal{C}_2 \mathcal{C}_2^*)^{-1}$.

We now describe several ways to obtain a state space realization for a given multiple input and multiple output transfer matrix $G(s)$. The simplest and most straightforward way to obtain a realization is by realizing each element of the matrix $G(s)$ and then combining all these individual realizations to form a realization for $G(s)$. To illustrate, let us consider a 2×2 (block) transfer matrix such as

$$G(s) = \begin{bmatrix} G_1(s) & G_2(s) \\ G_3(s) & G_4(s) \end{bmatrix}$$

and assume that $G_i(s)$ has a state space realization of

$$G_i(s) = \left[\begin{array}{c|c} A_i & B_i \\ \hline C_i & D_i \end{array}\right], \quad i = 1, \ldots, 4.$$

Note that $G_i(s)$ may itself be a multiple input and multiple output transfer matrix. In particular, if $G_i(s)$ is a column or row vector of transfer functions, then the formulas in Section 3.4 can be used to obtain a controllable or observable realization for $G_i(s)$. Then a realization for $G(s)$ can be given by

$$G(s) = \left[\begin{array}{cccc|cc} A_1 & 0 & 0 & 0 & B_1 & 0 \\ 0 & A_2 & 0 & 0 & 0 & B_2 \\ 0 & 0 & A_3 & 0 & B_3 & 0 \\ 0 & 0 & 0 & A_4 & 0 & B_4 \\ \hline C_1 & C_2 & 0 & 0 & D_1 & D_2 \\ 0 & 0 & C_3 & C_4 & D_3 & D_4 \end{array}\right].$$

Alternatively, if the transfer matrix $G(s)$ can be factored into the product and/or the sum of several simply realized transfer matrices, then a realization for G can be obtained by using the cascade or addition formulas in the last section.

A problem inherited with these kinds of realization procedures is that a realization thus obtained will generally not be minimal. To obtain a minimal realization, a Kalman controllability and observability decomposition has to be performed to eliminate the uncontrollable and/or unobservable states. (An alternative numerically reliable method to eliminate uncontrollable and/or unobservable states is the *balanced realization* method which will be discussed later.)

We will now describe one factorization procedure that does result in a minimal realization by using partial fractional expansion (The resulting realization is sometimes called *Gilbert's realization* due to Gilbert).

Let $G(s)$ be a $p \times m$ transfer matrix and write it in the following form:

$$G(s) = \frac{N(s)}{d(s)}$$

with $d(s)$ a scalar polynomial. For simplicity, we shall assume that $d(s)$ has only real and distinct roots $\lambda_i \neq \lambda_j$ if $i \neq j$ and

$$d(s) = (s-\lambda_1)(s-\lambda_2)\cdots(s-\lambda_r).$$

Then $G(s)$ has the following partial fractional expansion:

$$G(s) = D + \sum_{i=1}^{r} \frac{W_i}{s-\lambda_i}.$$

Suppose

$$\operatorname{rank} W_i = k_i$$

and let $B_i \in \mathbb{R}^{k_i \times m}$ and $C_i \in \mathbb{R}^{p \times k_i}$ be two constant matrices such that

$$W_i = C_i B_i.$$

Then a realization for $G(s)$ is given by

$$G(s) = \left[\begin{array}{ccc|c} \lambda_1 I_{k_1} & & & B_1 \\ & \ddots & & \vdots \\ & & \lambda_r I_{k_r} & B_r \\ \hline C_1 & \cdots & C_r & D \end{array}\right].$$

It follows immediately from PBH tests that this realization is controllable and observable. Hence, it is minimal.

An immediate consequence of this minimal realization is that a transfer matrix with an r-th order polynomial denominator does not necessarily have an r-th order state space realization unless W_i for each i is a rank one matrix.

This approach can, in fact, be generalized to more complicated cases where $d(s)$ may have complex and/or repeated roots. Readers may convince themselves by trying some simple examples.

3.8 Lyapunov Equations

Testing stability, controllability, and observability of a system is very important in linear system analysis and synthesis. However, these tests often have to be done indirectly. In that respect, the Lyapunov theory is sometimes useful. Consider the following Lyapunov equation

$$A^* X + XA + Q = 0 \tag{3.8}$$

with given real matrices A and Q. It has been shown in Chapter 2 that this equation has a unique solution iff $\lambda_i(A) + \bar{\lambda}_j(A) \neq 0, \forall i, j$. In this section, we will study the relationships between the stability of A and the solution of X. The following results are standard.

Lemma 3.18 *Assume that A is stable, then the following statements hold:*

(i) $X = \int_0^\infty e^{A^* t} Q e^{At} dt$.

(ii) $X > 0$ *if* $Q > 0$ *and* $X \geq 0$ *if* $Q \geq 0$.

(iii) if $Q \geq 0$, *then* (Q, A) *is observable iff* $X > 0$.

An immediate consequence of part (iii) is that, given a stable matrix A, a pair (C, A) is observable if and only if the solution to the following Lyapunov equation is positive definite:

$$A^* L_o + L_o A + C^* C = 0.$$

The solution L_o is called the *observability Gramian*. Similarly, a pair (A, B) is controllable if and only if the solution to

$$AL_c + L_c A^* + BB^* = 0$$

is positive definite and L_c is called the *controllability Gramian*.

In many applications, we are given the solution of the Lyapunov equation and need to conclude the stability of the matrix A.

Lemma 3.19 *Suppose X is the solution of the Lyapunov equation (3.8), then*

(i) $\mathrm{Re}\lambda_i(A) \leq 0$ if $X > 0$ and $Q \geq 0$.

(ii) A is stable if $X > 0$ and $Q > 0$.

(iii) A is stable if $X \geq 0$, $Q \geq 0$ and (Q, A) is detectable.

Proof. Let λ be an eigenvalue of A and $v \neq 0$ be a corresponding eigenvector, then $Av = \lambda v$. Pre-multiply equation (3.8) by v^* and postmultiply (3.8) by v to get

$$2\mathrm{Re}\ \lambda(v^* X v) + v^* Q v = 0.$$

Now if $X > 0$ then $v^* X v > 0$, and it is clear that $\mathrm{Re}\lambda \leq 0$ if $Q \geq 0$ and $\mathrm{Re}\lambda < 0$ if $Q > 0$. Hence (i) and (ii) hold. To see (iii), we assume $\mathrm{Re}\lambda \geq 0$. Then we must have $v^* Q v = 0$, i.e., $Qv = 0$. This implies that λ is an unstable and unobservable mode, which contradicts the assumption that (Q, A) is detectable. □

3.9 Balanced Realizations

Although there are infinitely many different state space realizations for a given transfer matrix, some particular realizations have proven to be very useful in control engineering and signal processing. Here we will only introduce one class of realizations for stable transfer matrices that are most useful in control applications. To motivate the class of realizations, we first consider some simple facts.

Lemma 3.20 *Let $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ be a state space realization of a (not necessarily stable) transfer matrix $G(s)$. Suppose that there exists a symmetric matrix*

$$P = P^* = \begin{bmatrix} P_1 & 0 \\ 0 & 0 \end{bmatrix}$$

with P_1 nonsingular such that

$$AP + PA^* + BB^* = 0.$$

Now partition the realization (A, B, C, D) compatibly with P as

$$\left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right].$$

Then

$$\left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is also a realization of G. *Moreover,* (A_{11}, B_1) *is controllable if* A_{11} *is stable.*

Proof. Use the partitioned P and (A, B, C) to get

$$0 = AP + PA^* + BB^* = \begin{bmatrix} A_{11}P_1 + P_1A_{11}^* + B_1B_1^* & P_1A_{21}^* + B_1B_2^* \\ A_{21}P_1 + B_2B_1^* & B_2B_2^* \end{bmatrix},$$

which gives $B_2 = 0$ and $A_{21} = 0$ since P_1 is nonsingular. Hence, part of the realization is not controllable:

$$\left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right] = \left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ 0 & A_{22} & 0 \\ \hline C_1 & C_2 & D \end{array}\right] = \left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right].$$

Finally, it follows from Lemma 3.18 that (A_{11}, B_1) is controllable if A_{11} is stable. $\square$

We also have

Lemma 3.21 *Let* $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *be a state space realization of a (not necessarily stable) transfer matrix* $G(s)$. *Suppose that there exists a symmetric matrix*

$$Q = Q^* = \begin{bmatrix} Q_1 & 0 \\ 0 & 0 \end{bmatrix}$$

with Q_1 *nonsingular such that*

$$QA + A^*Q + C^*C = 0.$$

Now partition the realization (A, B, C, D) *compatibly with* Q *as*

$$\left[\begin{array}{cc|c} A_{11} & A_{12} & B_1 \\ A_{21} & A_{22} & B_2 \\ \hline C_1 & C_2 & D \end{array}\right].$$

Then

$$\left[\begin{array}{c|c} A_{11} & B_1 \\ \hline C_1 & D \end{array}\right]$$

is also a realization of G. *Moreover,* (C_1, A_{11}) *is observable if* A_{11} *is stable.*

The above two lemmas suggest that to obtain a minimal realization from a stable non-minimal realization, one only needs to eliminate all states corresponding to the zero block diagonal term of the controllability Gramian P and the observability Gramian Q. In the case where P is not block diagonal, the following procedure can be used to eliminate non-controllable subsystems:

1. Let $G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ be a stable realization.

2. Compute the controllability Gramian $P \geq 0$ from
$$AP + PA^* + BB^* = 0.$$

3. Diagonalize P to get $P = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \Lambda_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_1 & U_2 \end{bmatrix}^*$ with $\Lambda_1 > 0$ and $\begin{bmatrix} U_1 & U_2 \end{bmatrix}$ unitary.

4. Then $G(s) = \left[\begin{array}{c|c} U_1^* A U_1 & U_1^* B \\ \hline C U_1 & D \end{array}\right]$ is a controllable realization.

A dual procedure can also be applied to eliminate non-observable subsystems.

Now assume that $\Lambda_1 > 0$ is diagonal and $\Lambda_1 = \mathrm{diag}(\Lambda_{11}, \Lambda_{12})$ such that $\lambda_{max}(\Lambda_{12}) \ll \lambda_{min}(\Lambda_{11})$, then it is tempting to conclude that one can also discard those states corresponding to Λ_{12} without causing much error. However, this is not necessarily true as shown in the following example. Consider a stable transfer function

$$G(s) = \frac{3s + 18}{s^2 + 3s + 18}.$$

Then $G(s)$ has a state space realization given by

$$G(s) = \left[\begin{array}{cc|c} -1 & -4/\alpha & 1 \\ 4\alpha & -2 & 2\alpha \\ \hline -1 & 2/\alpha & 0 \end{array}\right]$$

where α is any nonzero number. It is easy to check that the controllability Gramian of the realization is given by

$$P = \begin{bmatrix} 0.5 & \\ & \alpha^2 \end{bmatrix}.$$

Since the last diagonal term of P can be made arbitrarily small by making α small, the controllability of the corresponding state can be made arbitrarily weak. If the state corresponding to the last diagonal term of P is removed, we get a transfer function

$$\hat{G} = \left[\begin{array}{c|c} -1 & 1 \\ \hline -1 & 0 \end{array}\right] = \frac{-1}{s+1}$$

which is not close to the original transfer function in any sense. The problem may be easily detected if one checks the observability Gramian Q, which is

$$Q = \begin{bmatrix} 0.5 & \\ & 1/\alpha^2 \end{bmatrix}.$$

Since $1/\alpha^2$ is very large if α is small, this shows that the state corresponding to the last diagonal term is strongly observable. This example shows that controllability (or observability) Gramian alone can not give an accurate indication of the dominance of the system states in the input/output behavior.

This motivates the introduction of a balanced realization which gives balanced Gramians for controllability and observability.

Suppose $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ is stable, i.e., A is stable. Let P and Q denote the controllability Gramian and observability Gramian, respectively. Then by Lemma 3.18, P and Q satisfy the following Lyapunov equations

$$AP + PA^* + BB^* = 0 \tag{3.9}$$

$$A^*Q + QA + C^*C = 0, \tag{3.10}$$

and $P \geq 0$, $Q \geq 0$. Furthermore, the pair (A, B) is controllable iff $P > 0$, and (C, A) is observable iff $Q > 0$.

Suppose the state is transformed by a nonsingular T to $\hat{x} = Tx$ to yield the realization

$$G = \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right] = \left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right].$$

Then the Gramians are transformed to $\hat{P} = TPT^*$ and $\hat{Q} = (T^{-1})^*QT^{-1}$. Note that $\hat{P}\hat{Q} = TPQT^{-1}$, and therefore the eigenvalues of the product of the Gramians are invariant under state transformation.

Consider the similarity transformation T which gives the eigenvector decomposition

$$PQ = T^{-1}\Lambda T, \quad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

Then the columns of T^{-1} are eigenvectors of PQ corresponding to the eigenvalues $\{\lambda_i\}$. Later, it will be shown that PQ has a real diagonal Jordan form and that $\Lambda \geq 0$, which are consequences of $P \geq 0$ and $Q \geq 0$.

Although the eigenvectors are not unique, in the case of a minimal realization they can always be chosen such that

$$\hat{P} = TPT^* = \Sigma,$$

$$\hat{Q} = (T^{-1})^*QT^{-1} = \Sigma,$$

where $\Sigma = \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$ and $\Sigma^2 = \Lambda$. This new realization with controllability and observability Gramians $\hat{P} = \hat{Q} = \Sigma$ will be referred to as a *balanced realization* (also called internally balanced realization). The decreasingly order numbers, $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_n \geq 0$, are called the *Hankel singular values* of the system.

More generally, if a realization of a stable system is not minimal, then there is a transformation such that the controllability and observability Gramians for the transformed realization are diagonal and the controllable and observable subsystem is balanced. This is a consequence of the following matrix fact.

Theorem 3.22 *Let P and Q be two positive semidefinite matrices. Then there exists a nonsingular matrix T such that*

$$TPT^* = \begin{bmatrix} \Sigma_1 & & & \\ & \Sigma_2 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}, \quad (T^{-1})^* Q T^{-1} = \begin{bmatrix} \Sigma_1 & & & \\ & 0 & & \\ & & \Sigma_3 & \\ & & & 0 \end{bmatrix},$$

respectively, with Σ_1, Σ_2, Σ_3 diagonal and positive definite.

Proof. Since P is a positive semidefinite matrix, there exists a transformation T_1 such that

$$T_1 P T_1^* = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}.$$

Now let

$$(T_1^*)^{-1} Q T_1^{-1} = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{12}^* & Q_{22} \end{bmatrix},$$

and there exists a unitary matrix U_1 such that

$$U_1 Q_{11} U_1^* = \begin{bmatrix} \Sigma_1^2 & 0 \\ 0 & 0 \end{bmatrix}, \quad \Sigma_1 > 0.$$

Let

$$(T_2^*)^{-1} = \begin{bmatrix} U_1 & 0 \\ 0 & I \end{bmatrix},$$

and then

$$(T_2^*)^{-1}(T_1^*)^{-1} Q T_1^{-1}(T_2)^{-1} = \begin{bmatrix} \Sigma_1^2 & 0 & \hat{Q}_{121} \\ 0 & 0 & \hat{Q}_{122} \\ \hat{Q}_{121}^* & \hat{Q}_{122}^* & Q_{22} \end{bmatrix}.$$

But $Q \geq 0$ implies $\hat{Q}_{122} = 0$. So now let

$$(T_3^*)^{-1} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\hat{Q}_{121}^* \Sigma_1^{-2} & 0 & I \end{bmatrix},$$

giving

$$(T_3^*)^{-1}(T_2^*)^{-1}(T_1^*)^{-1} Q T_1^{-1}(T_2)^{-1}(T_3)^{-1} = \begin{bmatrix} \Sigma_1^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & Q_{22} - \hat{Q}_{121}^* \Sigma_1^{-2} \hat{Q}_{121} \end{bmatrix}.$$

Next find a unitary matrix U_2 such that

$$U_2 (Q_{22} - \hat{Q}_{121}^* \Sigma_1^{-2} \hat{Q}_{121}) U_2^* = \begin{bmatrix} \Sigma_3 & 0 \\ 0 & 0 \end{bmatrix}, \quad \Sigma_3 > 0.$$

Define

$$(T_4^*)^{-1} = \begin{bmatrix} \Sigma_1^{-1/2} & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & U_2 \end{bmatrix}$$

and let

$$T = T_4 T_3 T_2 T_1.$$

Then

$$TPT^* = \begin{bmatrix} \Sigma_1 & & & \\ & \Sigma_2 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}, \quad (T^*)^{-1} Q T^{-1} = \begin{bmatrix} \Sigma_1 & & & \\ & 0 & & \\ & & \Sigma_3 & \\ & & & 0 \end{bmatrix}$$

with $\Sigma_2 = I$. □

Corollary 3.23 *The product of two positive semi-definite matrices is similar to a positive semi-definite matrix.*

Proof. Let P and Q be any positive semi-definite matrices. Then it is easy to see that with the transformation given above

$$TPQT^{-1} = \begin{bmatrix} \Sigma_1^2 & 0 \\ 0 & 0 \end{bmatrix}.$$

□

Corollary 3.24 *For any stable system* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$, *there exists a nonsingular* T *such that* $G = \left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right]$ *has controllability Gramian* P *and observability Gramian* Q *given by*

$$P = \begin{bmatrix} \Sigma_1 & & & \\ & \Sigma_2 & & \\ & & 0 & \\ & & & 0 \end{bmatrix}, \quad Q = \begin{bmatrix} \Sigma_1 & & & \\ & 0 & & \\ & & \Sigma_3 & \\ & & & 0 \end{bmatrix},$$

respectively, with Σ_1, Σ_2, Σ_3 *diagonal and positive definite.*

In the special case where $\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ is a minimal realization, a balanced realization can be obtained through the following simplified procedure:

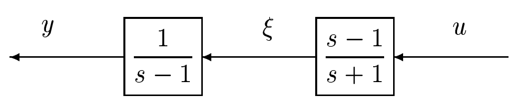
1. Compute the controllability and observability Gramians $P > 0, Q > 0$.
2. Find a matrix R such that $P = R^*R$.
3. Diagonalize RQR^* to get $RQR^* = U\Sigma^2U^*$.
4. Let $T^{-1} = R^*U\Sigma^{-1/2}$. Then $TPT^* = (T^*)^{-1}QT^{-1} = \Sigma$ and $\left[\begin{array}{c|c} TAT^{-1} & TB \\ \hline CT^{-1} & D \end{array}\right]$ is balanced.

Assume that the Hankel singular values of the system is decreasingly ordered so that $\Sigma = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_n)$ and $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_n$ and suppose $\sigma_r \gg \sigma_{r+1}$ for some r then the balanced realization implies that those states corresponding to the singular values of $\sigma_{r+1}, \ldots, \sigma_n$ are less controllable and observable than those states corresponding to $\sigma_1, \ldots, \sigma_r$. Therefore, truncating those less controllable and observable states will not lose much information about the system. These statements will be made more concrete in Chapter 7 where error bounds will be derived for the truncation error.

Two other closely related realizations are called *input normal realization* with $P = I$ and $Q = \Sigma^2$, and *output normal realization* with $P = \Sigma^2$ and $Q = I$. Both realizations can be obtained easily from the balanced realization by a suitable scaling on the states.

3.10 Hidden Modes and Pole-Zero Cancelation

Another important issue associated with the realization theory is the problem of uncontrollable and/or unobservable unstable modes in the dynamical system. This problem is illustrated in the following example: Consider a series connection of two subsystems as shown in the following diagram



The transfer function for this system,

$$g(s) = \frac{s-1}{s+1}\frac{1}{s-1} = \frac{1}{s+1},$$

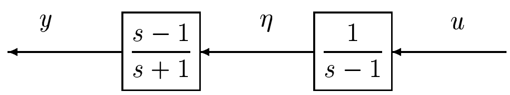
is stable and has a first order minimal realization. On the other hand, let

$$\begin{aligned} x_1 &= y \\ x_2 &= u - \xi. \end{aligned}$$

Then a state space description for this dynamical system is given by

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

and is a second order system. Moreover, it is easy to show that the unstable mode 1 is uncontrollable but observable. Hence, the output can be unbounded if the initial state $x_1(0)$ is not zero. We should also note that the above problem does not go away by changing the interconnection order:



In the later case, the unstable mode 1 becomes controllable but unobservable. The unstable mode can still result in the internal signal η unbounded if the initial state $\eta(0)$ is not zero. Of course, there are fundamental differences between these two types of interconnections as far as control design is concerned. For instance, if the state is available for feedback control, then the latter interconnection can be stabilized while the former cannot be.

This example shows that we must be very careful in canceling unstable modes in the procedure of forming a transfer function in control designs; otherwise the results obtained may be misleading and those unstable modes become hidden modes waiting to blow. One observation from this example is that the problem is really caused by the unstable zero of the subsystem $\frac{s-1}{s+1}$. Although the zeros of an SISO transfer function are easy to see, it is not quite so for an MIMO transfer matrix. In fact, the notion of "system zero" cannot be generalized naturally from the scalar transfer function zeros. For example, consider the following transfer matrix

$$G(s) = \begin{bmatrix} \dfrac{1}{s+1} & \dfrac{1}{s+2} \\ \dfrac{2}{s+2} & \dfrac{1}{s+1} \end{bmatrix}$$

which is stable and each element of $G(s)$ has no finite zeros. Let

$$K = \begin{bmatrix} \dfrac{s+2}{s-\sqrt{2}} & -\dfrac{s+1}{s-\sqrt{2}} \\ 0 & 1 \end{bmatrix}$$

which is unstable. However,

$$KG = \begin{bmatrix} -\dfrac{s+\sqrt{2}}{(s+1)(s+2)} & 0 \\ \dfrac{2}{s+2} & \dfrac{1}{s+1} \end{bmatrix}$$

is stable. This implies that $G(s)$ must have an unstable zero at $\sqrt{2}$ that cancels the unstable pole of K. This leads us to the next topic: multivariable system poles and zeros.

3.11 Multivariable System Poles and Zeros

Let $\mathbb{R}[s]$ denote the polynomial ring with real coefficients. A matrix is called a polynomial matrix if every element of the matrix is in $\mathbb{R}[s]$. A square polynomial matrix is called a *unimodular matrix* if its determinant is a nonzero constant (not a polynomial of s). It is a fact that a square polynomial matrix is unimodular if and only if it is invertible in $\mathbb{R}[s]$, i.e., its inverse is also a polynomial matrix.

Definition 3.11 Let $Q(s) \in \mathbb{R}[s]$ be a $(p \times m)$ polynomial matrix. Then the *normal rank* of $Q(s)$, denoted *normalrank* $(Q(s))$, is the rank in $\mathbb{R}[s]$ or, equivalently, is the maximum dimension of a square submatrix of $Q(s)$ with nonzero determinant in $\mathbb{R}[s]$.

In short, sometimes we say that a polynomial matrix $Q(s)$ has rank$(Q(s))$ in $\mathbb{R}[s]$ when we refer to the normal rank of $Q(s)$.

To show the difference between the normal rank of a polynomial matrix and the rank of the polynomial matrix evaluated at certain point, consider

$$Q(s) = \begin{bmatrix} s & 1 \\ s^2 & 1 \end{bmatrix}.$$

Then $Q(s)$ has normal rank 2 since $\det Q(s) = s - s^2 \not\equiv 0$. However, $Q(0)$ has rank 1.

It is a fact in linear algebra that any polynomial matrix can be reduced to a so-called *Smith form* through some pre- and post- unimodular operations. [cf. Kailath, 1984, pp.391].

Lemma 3.25 (Smith form) *Let $P(s) \in \mathbb{R}[s]$ be any polynomial matrix, then there exist unimodular matrices $U(s), V(s) \in \mathbb{R}[s]$ such that*

$$U(s)P(s)V(s) = S(s) := \begin{bmatrix} \gamma_1(s) & 0 & \cdots & 0 & 0 \\ 0 & \gamma_2(s) & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \gamma_r(s) & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

and $\gamma_i(s)$ divides $\gamma_{i+1}(s)$.

$S(s)$ is called the *Smith form* of $P(s)$. It is also clear that r is the normal rank of $P(s)$. We shall illustrate the procedure of obtaining a Smith form by an example. Let

$$P(s) = \begin{bmatrix} s+1 & (s+1)(2s+1) & s(s+1) \\ s+2 & (s+2)(s^2+5s+3) & s(s+2) \\ 1 & 2s+1 & s \end{bmatrix}.$$

The polynomial matrix $P(s)$ has normal rank 2 since

$$\det(P(s)) \equiv 0, \quad \det \begin{bmatrix} s+1 & (s+1)(2s+1) \\ s+2 & (s+2)(s^2+5s+3) \end{bmatrix} = (s+1)^2(s+2)^2 \not\equiv 0.$$

First interchange the first row and the third row and use row elementary operation to zero out the $s+1$ and $s+2$ elements of $P(s)$. This process can be done by pre-multiplying a unimodular matrix U to $P(s)$:

$$U = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -(s+2) \\ 1 & 0 & -(s+1) \end{bmatrix}.$$

Then

$$P_1(s) := U(s)P(s) = \begin{bmatrix} 1 & 2s+1 & s \\ 0 & (s+1)(s+2)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Next use column operation to zero out the $2s+1$ and s terms in P_1. This process can be done by post-multiplying a unimodular matrix V to $P_1(s)$:

$$V(s) = \begin{bmatrix} 1 & -(2s+1) & -s \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and

$$P_1(s)V(s) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (s+1)(s+2)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then we have

$$S(s) = U(s)P(s)V(s) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & (s+1)(s+2)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Similarly, let $\mathcal{R}_p(s)$ denote the set of rational proper transfer matrices.[1] Then any real rational transfer matrix can be reduced to a so-called *McMillan form* through some pre- and post- unimodular operations.

Lemma 3.26 (McMillan form) *Let $G(s) \in \mathcal{R}_p(s)$ be any proper real rational transfer matrix, then there exist unimodular matrices $U(s), V(s) \in \mathbb{R}[s]$ such that*

$$U(s)G(s)V(s) = M(s) := \begin{bmatrix} \frac{\alpha_1(s)}{\beta_1(s)} & 0 & \cdots & 0 & 0 \\ 0 & \frac{\alpha_2(s)}{\beta_2(s)} & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \frac{\alpha_r(s)}{\beta_r(s)} & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

and $\alpha_i(s)$ divides $\alpha_{i+1}(s)$ and $\beta_{i+1}(s)$ divides $\beta_i(s)$.

[1] It is obvious that a similar notion of normal rank can be extended to a matrix in $\mathcal{R}_p(s)$. In particular, let $G(s) \in \mathcal{R}_p(s)$ be a rational matrix and write $G(s) = N(s)/d(s)$ such that $d(s)$ is a polynomial and $N(s)$ is a polynomial matrix, then $G(s)$ is said to have normal rank r if $N(s)$ has normal rank r.

Proof. If we write the transfer matrix $G(s)$ as $G(s) = N(s)/d(s)$ such that $d(s)$ is a scalar polynomial and $N(s)$ is a $p \times m$ polynomial matrix and if let the Smith form of $N(s)$ be $S(s) = U(s)N(s)V(s)$, the conclusion follows by letting $M(s) = S(s)/d(s)$. □

Definition 3.12 The number $\sum_i \deg(\beta_i(s))$ is called the *McMillan degree* of $G(s)$ where $\deg(\beta_i(s))$ denotes the degree of the polynomial $\beta_i(s)$, i.e., the highest power of s in $\beta_i(s)$.

The McMillan degree of a transfer matrix is closely related to the dimension of a minimal realization of $G(s)$. In fact, it can be shown that the dimension of a minimal realization of $G(s)$ is exactly the McMillan degree of $G(s)$.

Definition 3.13 The roots of all the polynomials $\beta_i(s)$ in the McMillan form for $G(s)$ are called the *poles* of G.

Let (A, B, C, D) be a minimal realization of $G(s)$. Then it is fairly easy to show that a complex number is a pole of $G(s)$ if and only if it is an eigenvalue of A.

Definition 3.14 The roots of all the polynomials $\alpha_i(s)$ in the McMillan form for $G(s)$ are called the *transmission zeros* of $G(s)$. A complex number $z_0 \in \mathbb{C}$ is called a *blocking zero* of $G(s)$ if $G(z_0) = 0$.

It is clear that a blocking zero is a transmission zero. Moreover, for a scalar transfer function, the blocking zeros and the transmission zeros are the same.

We now illustrate the above concepts through an example. Consider a 3×3 transfer matrix:

$$G(s) = \begin{bmatrix} \dfrac{1}{(s+1)(s+2)} & \dfrac{2s+1}{(s+1)(s+2)} & \dfrac{s}{(s+1)(s+2)} \\ \dfrac{1}{(s+1)^2} & \dfrac{s^2+5s+3}{(s+1)^2} & \dfrac{s}{(s+1)^2} \\ \dfrac{1}{(s+1)^2(s+2)} & \dfrac{2s+1}{(s+1)^2(s+2)} & \dfrac{s}{(s+1)^2(s+2)} \end{bmatrix}.$$

Then $G(s)$ can be written as

$$G(s) = \frac{1}{(s+1)^2(s+2)} \begin{bmatrix} s+1 & (s+1)(2s+1) & s(s+1) \\ s+2 & (s+2)(s^2+5s+3) & s(s+2) \\ 1 & 2s+1 & s \end{bmatrix} := \frac{N(s)}{d(s)}.$$

Since $N(s)$ is exactly the same as the $P(s)$ in the previous example, it is clear that the $G(s)$ has the McMillan form

$$M(s) = U(s)G(s)V(s) = \begin{bmatrix} \frac{1}{(s+1)^2(s+2)} & 0 & 0 \\ 0 & \frac{s+2}{s+1} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and $G(s)$ has McMillan degree of 4. The poles of the transfer matrix are $\{-1, -1, -1, -2\}$ and the transmission zero is $\{-2\}$. Note that the transfer matrix has pole and zero at the same location $\{-2\}$; this is the unique feature of multivariable systems.

To get a minimal state space realization for $G(s)$, note that $G(s)$ has the following partial fractional expansion:

$$\begin{aligned} G(s) \quad = \quad & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \frac{1}{s+1}\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} 1 & -1 & -1 \end{bmatrix} + \frac{1}{s+1}\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\begin{bmatrix} 0 & 3 & 1 \end{bmatrix} \\ & + \frac{1}{s+1}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 3 & 2 \end{bmatrix} + \frac{1}{(s+1)^2}\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & -1 \end{bmatrix} \\ & + \frac{1}{s+2}\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & -3 & -2 \end{bmatrix} \end{aligned}$$

Since there are repeated poles at -1, the Gilbert's realization procedure described in the last section cannot be used directly. Nevertheless, a careful inspection of the fractional expansion results in a 4-th order minimal state space realization:

$$G(s) = \left[\begin{array}{cccc|ccc} -1 & 0 & 1 & 0 & 0 & 3 & 1 \\ 0 & -1 & 1 & 0 & -1 & 3 & 2 \\ 0 & 0 & -1 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & -2 & 1 & -3 & -2 \\ \hline 0 & 0 & 1 & -1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \end{array}\right].$$

We remind readers that there are many different definitions of system zeros. The definitions introduced here are the most common ones and are useful in this book.

Lemma 3.27 *Let $G(s)$ be a $p \times m$ proper transfer matrix with full column normal rank. Then $z_0 \in \mathbb{C}$ is a transmission zero of $G(s)$ if and only if there exists a $0 \neq u_0 \in \mathbb{C}^m$ such that $G(z_0)u_0 = 0$.*

Proof. We shall outline a proof of this lemma. We shall first show that there is a vector $u_0 \in \mathbb{C}^m$ such that $G(z_0)u_0 = 0$ if $z_0 \in \mathbb{C}$ is a transmission zero. Without loss of generality, assume

$$G(s) = U_1(s) \begin{bmatrix} \frac{\alpha_1(s)}{\beta_1(s)} & 0 & \cdots & 0 \\ 0 & \frac{\alpha_2(s)}{\beta_2(s)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\alpha_m(s)}{\beta_m(s)} \\ 0 & 0 & \cdots & 0 \end{bmatrix} V_1(s)$$

for some unimodular matrices $U_1(s)$ and $V_1(s)$ and suppose z_0 is a zero of $\alpha_1(s)$, i.e., $\alpha_1(z_0) = 0$. Let

$$u_0 = V_1^{-1}(z_0)e_1 \neq 0$$

where $e_1 = [1, 0, 0, \ldots]^* \in \mathbb{R}^m$. Then it is easy to verify that $G(z_0)u_0 = 0$. On the other hand, suppose there is a $u_0 \in \mathbb{C}^m$ such that $G(z_0)u_0 = 0$. Then

$$U_1(z_0) \begin{bmatrix} \frac{\alpha_1(z_0)}{\beta_1(z_0)} & 0 & \cdots & 0 \\ 0 & \frac{\alpha_2(z_0)}{\beta_2(z_0)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{\alpha_m(z_0)}{\beta_m(z_0)} \\ 0 & 0 & \cdots & 0 \end{bmatrix} V_1(z_0)u_0 = 0.$$

Define

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} = V_1(z_0)u_0 \neq 0.$$

Then

$$\begin{bmatrix} \alpha_1(z_0)u_1 \\ \alpha_2(z_0)u_2 \\ \vdots \\ \alpha_m(z_0)u_m \end{bmatrix} = 0.$$

This implies that z_0 must be a root of one of polynomials $\alpha_i(s), i = 1, \ldots, m$. $\square$

Note that the lemma may not be true if $G(s)$ does not have full column normal rank. This can be seen from the following example. Consider

$$G(s) = \frac{1}{s+1} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad u_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

It is easy to see that G has no transmission zero but $G(s)u_0 = 0$ for all s. It should also be noted that the above lemma applies even if z_0 is a pole of $G(s)$ although $G(z_0)$ is not defined. The reason is that $G(z_0)u_0$ may be well defined. For example,

$$G(s) = \begin{bmatrix} \frac{s-1}{s+1} & 0 \\ 0 & \frac{s+2}{s-1} \end{bmatrix}, \quad u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then $G(1)u_0 = 0$. Therefore, 1 is a transmission zero.

Similarly we have the following lemma:

Lemma 3.28 *Let $G(s)$ be a $p \times m$ proper transfer matrix with full row normal rank. Then $z_0 \in \mathbb{C}$ is a transmission zero of $G(s)$ if and only if there exists a $0 \neq \eta_0 \in \mathbb{C}^p$ such that $\eta_0^* G(z_0) = 0$.*

In the case where the transmission zero is not a pole of $G(s)$, we can give a useful alternative characterization of the transfer matrix transmission zeros. Furthermore, $G(s)$ is not required to be full column (or row) rank in this case.

The following lemma is easy to show from the definition of zeros.

Lemma 3.29 *Suppose $z_0 \in \mathbb{C}$ is not a pole of $G(s)$. Then z_0 is a transmission zero if and only if* $\operatorname{rank}(G(z_0)) < \operatorname{normalrank}(G(s))$.

Corollary 3.30 *Let $G(s)$ be a square $m \times m$ proper transfer matrix and* $\det G(s) \not\equiv 0$. *Suppose $z_o \in \mathbb{C}$ is not a pole of $G(s)$. Then $z_0 \in \mathbb{C}$ is a transmission zero of $G(s)$ if and only if* $\det G(z_0) = 0$.

Using the above corollary, we can confirm that the example in the last section does have a zero at $\sqrt{2}$ since

$$\det \begin{bmatrix} \dfrac{1}{s+1} & \dfrac{1}{s+2} \\ \dfrac{2}{s+2} & \dfrac{1}{s+1} \end{bmatrix} = \frac{2 - s^2}{(s+1)^2(s+2)^2}.$$

Note that the above corollary may not be true if z_0 is a pole of G. For example,

$$G(s) = \begin{bmatrix} \frac{s-1}{s+1} & 0 \\ 0 & \frac{s+2}{s-1} \end{bmatrix}$$

has a zero at 1 which is not a zero of $\det G(s)$.

The poles and zeros of a transfer matrix can also be characterized in terms of its state space realizations. Let

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$$

be a state space realization of $G(s)$.

Definition 3.15 The eigenvalues of A are called the *poles* of the realization of $G(s)$.

To define zeros, let us consider the following system matrix

$$Q(s) = \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}.$$

Definition 3.16 A complex number $z_0 \in \mathbb{C}$ is called an *invariant zero* of the system realization if it satisfies

$$\operatorname{rank} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} < normalrank \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}.$$

The invariant zeros are not changed by constant state feedback since

$$\operatorname{rank} \begin{bmatrix} A + BF - z_0 I & B \\ C + DF & D \end{bmatrix} = \operatorname{rank} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} I & 0 \\ F & I \end{bmatrix} = \operatorname{rank} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix}.$$

It is also clear that invariant zeros are not changed under similarity transformation.

The following lemma is obvious.

Lemma 3.31 *Suppose* $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$ *has full column normal rank. Then* $z_0 \in \mathbb{C}$ *is an invariant zero of a realization* (A, B, C, D) *if and only if there exist* $0 \neq x \in \mathbb{C}^n$ *and* $u \in \mathbb{C}^m$ *such that*

$$\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = 0.$$

Moreover, if $u = 0$*, then* z_0 *is also a non-observable mode.*

Proof. By definition, z_0 is an invariant zero if there is a vector $\begin{bmatrix} x \\ u \end{bmatrix} \neq 0$ such that

$$\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = 0$$

since $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$ has full column normal rank.

On the other hand, suppose z_0 is an invariant zero, then there is a vector $\begin{bmatrix} x \\ u \end{bmatrix} \neq 0$ such that

$$\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = 0.$$

We claim that $x \neq 0$. Otherwise, $\begin{bmatrix} B \\ D \end{bmatrix} u = 0$ or $u = 0$ since $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$ has full column normal rank, i.e., $\begin{bmatrix} x \\ u \end{bmatrix} = 0$ which is a contradiction.

Finally, note that if $u = 0$, then

$$\begin{bmatrix} A - z_0 I \\ C \end{bmatrix} x = 0$$

and z_0 is a non-observable mode by PBH test. □

Lemma 3.32 *Suppose* $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$ *has full row normal rank. Then* $z_0 \in \mathbb{C}$ *is an invariant zero of a realization* (A, B, C, D) *if and only if there exist* $0 \neq y \in \mathbb{C}^n$ *and* $v \in \mathbb{C}^p$ *such that*

$$\begin{bmatrix} y^* & v^* \end{bmatrix} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} = 0.$$

Moreover, if $v = 0$, *then* z_0 *is also a non-controllable mode.*

Lemma 3.33 $G(s)$ *has full column (row) normal rank if and only if* $\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$ *has full column (row) normal rank.*

Proof. This follows by noting that

$$\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ C(A - sI)^{-1} & I \end{bmatrix} \begin{bmatrix} A - sI & B \\ 0 & G(s) \end{bmatrix}$$

and

$$\text{normalrank} \begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = n + \text{normalrank}(G(s)).$$

□

Theorem 3.34 *Let* $G(s)$ *be a real rational proper transfer matrix and let* (A, B, C, D) *be a corresponding minimal realization. Then a complex number* z_0 *is a transmission zero of* $G(s)$ *if and only if it is an invariant zero of the minimal realization.*

Proof. We will give a proof only for the case that the transmission zero is not a pole of $G(s)$. Then, of course, z_0 is not an eigenvalue of A since the realization is minimal. Note that

$$\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ C(A - sI)^{-1} & I \end{bmatrix} \begin{bmatrix} A - sI & B \\ 0 & G(s) \end{bmatrix}.$$

Since we have assumed that z_0 is not a pole of $G(s)$, we have

$$\text{rank} \begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} = n + \text{rank } G(z_0).$$

Hence

$$\operatorname{rank}\begin{bmatrix} A - z_0 I & B \\ C & D \end{bmatrix} < \operatorname{normalrank}\begin{bmatrix} A - sI & B \\ C & D \end{bmatrix}$$

if and only if rank $G(z_0) <$ normalrank $G(s)$. Then the conclusion follows from Lemma 3.29. □

Note that the minimality assumption is essential for the converse statement. For example, consider a transfer matrix $G(s) = D$ (constant) and a realization of $G(s) = \left[\begin{array}{c|c} A & 0 \\ \hline C & D \end{array}\right]$ where A is any square matrix with any dimension and C is any matrix with compatible dimension. Then $G(s)$ has no poles or zeros but every eigenvalue of A is an invariant zero of the realization $\left[\begin{array}{c|c} A & 0 \\ \hline C & D \end{array}\right]$.

Nevertheless, we have the following corollary if a realization of the transfer matrix is not minimal.

Corollary 3.35 *Every transmission zero of a transfer matrix $G(s)$ is an invariant zero of all its realizations, and every pole of a transfer matrix $G(s)$ is a pole of all its realizations.*

Lemma 3.36 *Let $G(s) \in \mathcal{R}_p(s)$ be a $p \times m$ transfer matrix and let (A, B, C, D) be a minimal realization. If the system input is of the form $u(t) = u_0 e^{\lambda t}$, where $\lambda \in \mathbb{C}$ is not a pole of $G(s)$ and $u_0 \in \mathbb{C}^m$ is an arbitrary constant vector, then the output due to the input $u(t)$ and the initial state $x(0) = (\lambda I - A)^{-1} B u_0$ is $y(t) = G(\lambda) u_0 e^{\lambda t}$, $\forall t \geq 0$.*

Proof. The system response with respect to the input $u(t) = u_0 e^{\lambda t}$ and the initial condition $x(0) = (\lambda I - A)^{-1} B u_0$ is (in terms of Laplace transform):

$$\begin{aligned} Y(s) &= C(sI - A)^{-1} x(0) + C(sI - A)^{-1} B U(s) + D U(s) \\ &= C(sI - A)^{-1} x(0) + C(sI - A)^{-1} B u_0 (s - \lambda)^{-1} + D u_0 (s - \lambda)^{-1} \\ &= C(sI - A)^{-1} (x(0) - (\lambda I - A)^{-1} B u_0) + G(\lambda) u_0 (s - \lambda)^{-1} \\ &= G(\lambda) u_0 (s - \lambda)^{-1}. \end{aligned}$$

Hence $y(t) = G(\lambda) u_0 e^{\lambda t}$. □

Combining the above two lemmas, we have the following results that give a dynamical interpretation of a system's transmission zero.

Corollary 3.37 *Let $G(s) \in \mathcal{R}_p(s)$ be a $p \times m$ transfer matrix and let (A, B, C, D) be a minimal realization. Suppose that $z_0 \in \mathbb{C}$ is a transmission zero of $G(s)$ and is not a pole of $G(s)$. Then for any nonzero vector $u_0 \in \mathbb{C}^m$ the output of the system due to the initial state $x(0) = (z_0 I - A)^{-1} B u_0$ and the input $u = u_0 e^{z_0 t}$ is identically zero: $y(t) = G(z_0) u_0 e^{z_0 t} = 0$.*

The following lemma characterizes the relationship between zeros of a transfer function and poles of its inverse.

Lemma 3.38 *Suppose that* $G = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ *is a square transfer matrix with* D *nonsingular, and suppose* z_0 *is not an eigenvalue of* A *(note that the realization is not necessarily minimal). Then there exists* x_0 *such that*

$$(A - BD^{-1}C)x_0 = z_0 x_0, \quad Cx_0 \neq 0$$

iff there exists $u_0 \neq 0$ *such that*

$$G(z_0)u_0 = 0.$$

Proof. ($\Leftarrow$) $G(z_0)u_0 = 0$ implies that

$$G^{-1}(s) = \left[\begin{array}{c|c} A - BD^{-1}C & -BD^{-1} \\ \hline D^{-1}C & D^{-1} \end{array}\right]$$

has a pole at z_0 which is observable. Then, by definition, there exists x_0 such that

$$(A - BD^{-1}C)x_0 = z_0 x_0$$

and

$$Cx_0 \neq 0.$$

($\Rightarrow$) Set $u_0 = -D^{-1}Cx_0 \neq 0$. Then

$$(z_0 I - A)x_0 = -BD^{-1}Cx_0 = Bu_0.$$

Using this equality, one gets

$$G(z_0)u_0 = C(z_0 I - A)^{-1}Bu_0 + Du_0 = Cx_0 - Cx_0 = 0.$$

□

The above lemma implies that z_0 is a zero of an invertible $G(s)$ if and only if it is a pole of $G^{-1}(s)$.

3.12 Notes and References

Readers are referred to Brogan [1991], Chen [1984], Kailath [1980], and Wonham [1985] for the extensive treatment of the standard linear system theory. The balanced realization was first introduced by Mullis and Roberts [1976] to study the roundoff noise in digital filters. Moore [1981] proposed the balanced truncation method for model reduction which will be considered in Chapter 7.